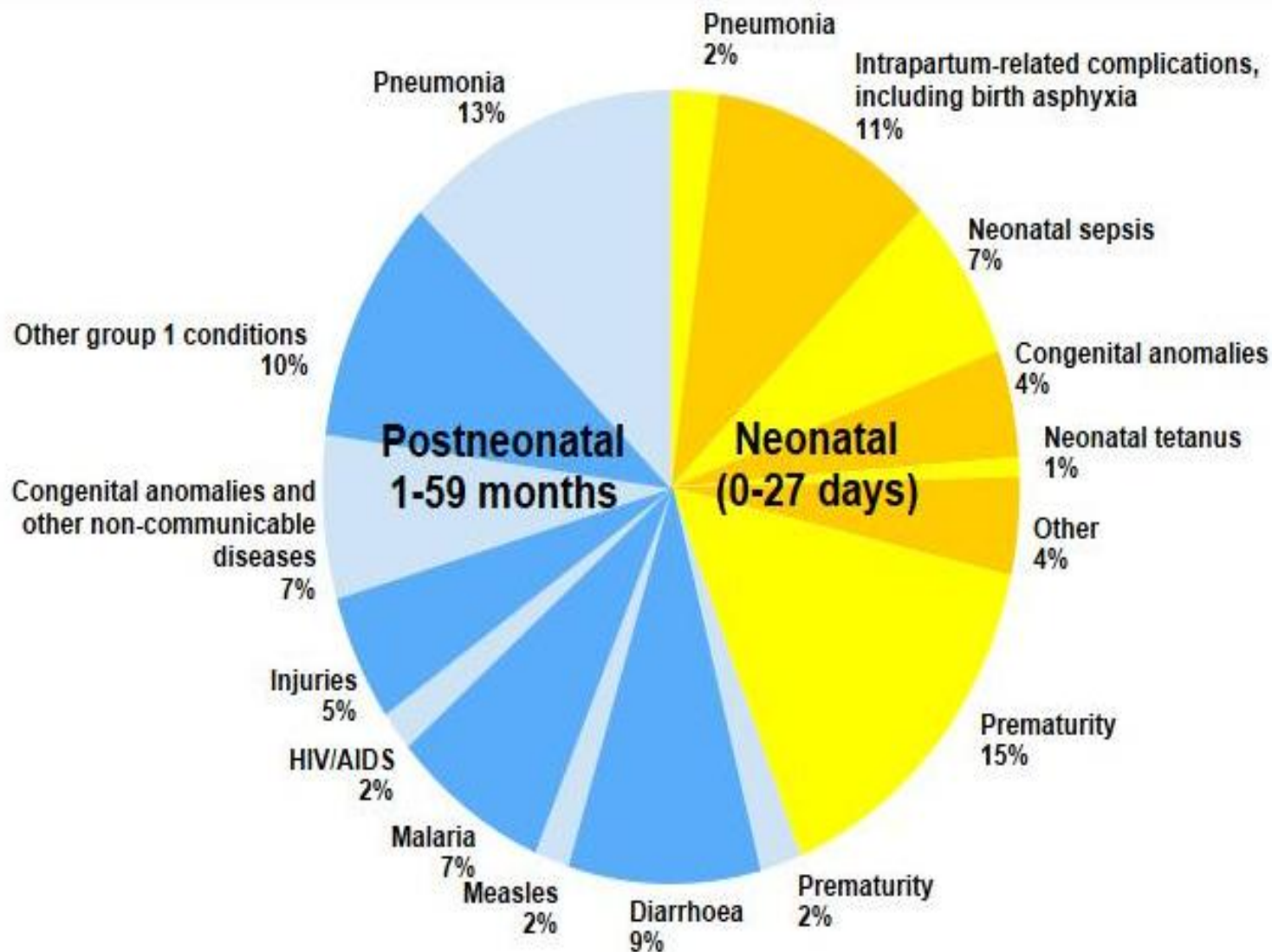


CHAPTER :5

Neonatology

- Neo –new , nata –birth , logy –study
- **Neonatology** is a subspecialty of pediatrics that deals with medical care of newborn infants (neonate)
- **Neonate** : birth to 28 days(less than 4 wks age).
- **A Neonatal Intensive Care Unit (NICU)** is a special area of the hospital that is devoted to the care of critically ill neonate.

Causes of deaths among children under 5 years, 2013



Overview of neonatal health...

- **Ethiopia**, Every year, more than 257,000 children under the age of five die and
- 120,000 die in the neonatal period.
- >40 percent of under-five deaths in Ethiopia are neonatal deaths.
- 2.4 million died in the first month of life(world)
- **Neonatal mortality rate** – 33 (in 2019)

Overview of neonatal health...

❖ *Major cause of death for neonate in Ethiopia are:*

- *neonatal infection (36%),*
- *birth asphyxia (25%),*
- preterm (17%),
- tetanus (7%),
- congenital problem (4%) and
- diarrhoea (3%).

Newborn assesement

HISTORY TAKING FOR NEONATE

- 1. Identification data:** name, age, sex, address, birth date, admission date
- 2.Source of information:** commonly mother
- 3.Chief complain:** the reason why comes to the health institution with duration .
- 4.History of present illness:** Chronologic narrative of the chief complaint.

HISTORY TAKING FOR NEONATE..

5. Obstetrics history:

- LMP, EDD (by dates/palpation/ultrasound),
- HIV test ,VDRL test , Blood group,
- ANC attendance
- any maternal and foetal problems
- medication/substance taken
- Previous Pregnancies problem and Outcomes.

HISTORY TAKING FOR NEONATE..

6. Labour and birth history:

- Onset and duration of labor,
- Rupture of membranes (mode and duration),
- Problems (maternal and foetal)
- birth complication, mode of delivery,
- fetal presentation, place of delivery

HISTORY TAKING FOR NEONATE..

7. Neonatal history:

- APGAR score, weight,
- resuscitation history,
- any neonatal problem ,
- breast feeding,
- time of discharged from health institution,
- postnatal care follow up.

HISTORY TAKING FOR NEONATE..

8. Maternal history:

- Age, gravid, para, abortion ,
- chronic illness (DM ,hypetension, heart disease, renal problem, epilepsy, cancer, HIV/AIDS....),
- infection,
- any medical/surgical problem ,
- medication/substance use
- education status,
- socioeconomic status, life style
- stress/anxiety.....

HISTORY TAKING FOR NEONATE..

9. Family history:

- history of familial related disease like (DM, Hypertension, Epileptic disorder, any cardiac , renal problems.....),
- any medical problem in the family,
- twins in family ,
- life style,
- family size,
- father health history,
- both mother and father live together or not.

PHYSICAL EXAMINATION FOR NEONATE

➤ During examining the neonate:

- Hand washing,
- don glove,
- Thermal environment,
- Light .

P/E FOR NEONATE..

1. General appearance-

- General body built,
- Muscle activity,
- Congenital anomalies,
- Respiratory effort,
- Acute sick looking or Chronic sick looking,
- Dysmorphic feature,
- Gross deformities,
- Syndromes

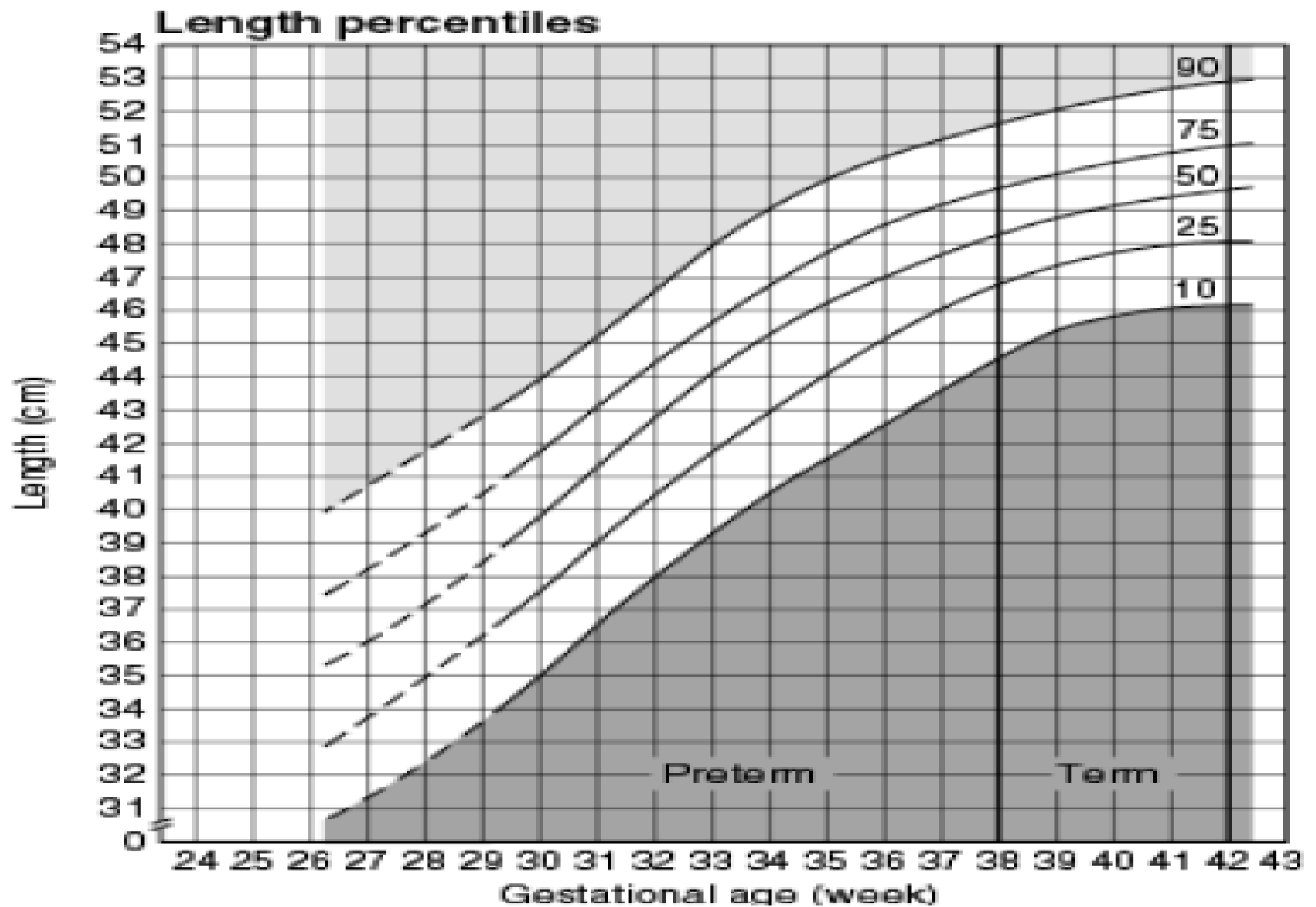
P/E FOR NEONATE..

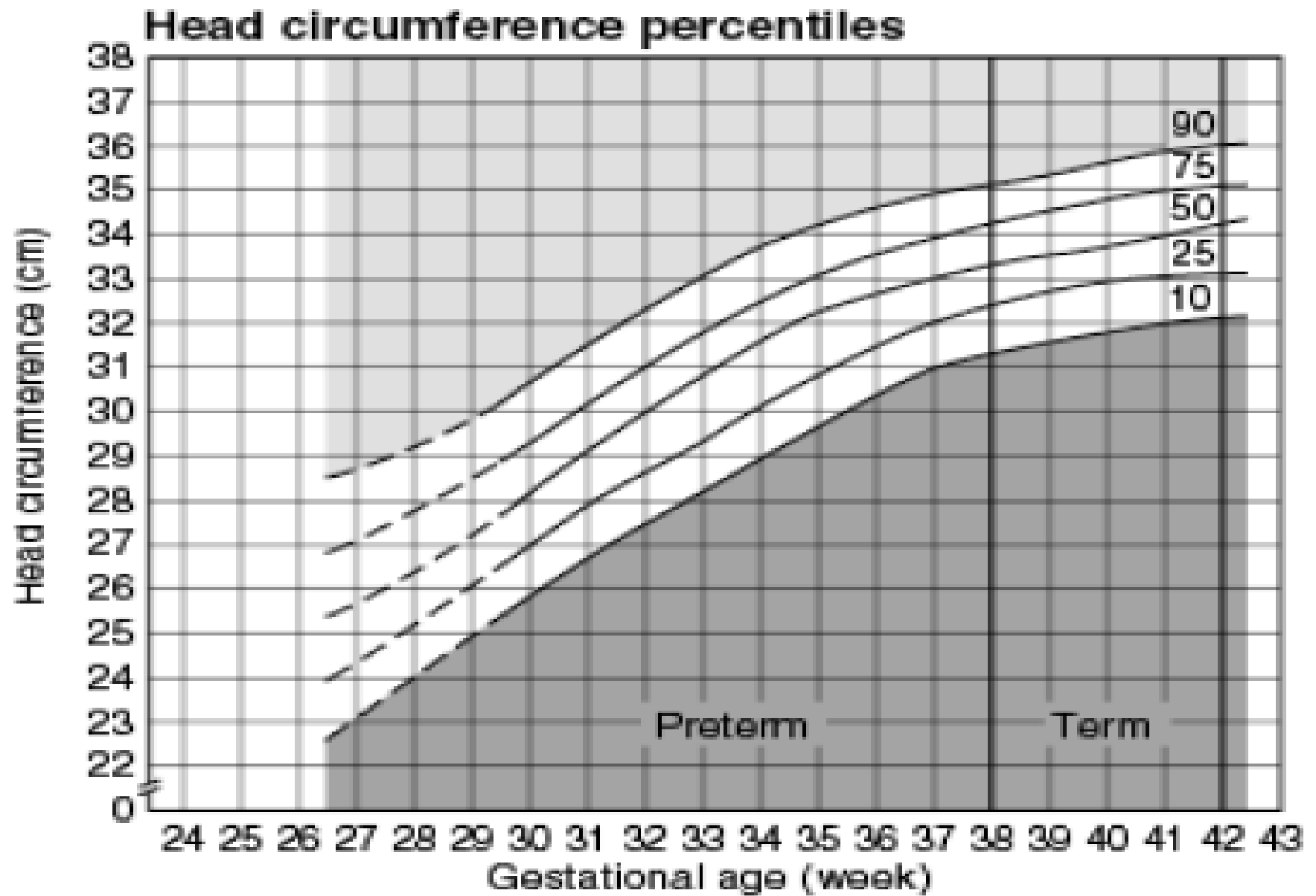
2. Vital sign-

- Temp, PR,
- RR, BP,
- oxygen saturation

3. Anthropometric:

- weight,
- length ,
- HC





P/E FOR NEONATE..

4. HEENT(head, eye, ear, nose, neck, throat):

➤ Head:

- Palpable sutures , *Molding* ,
- Masses from birth trauma,
- Caput succedaneum: cross suture line
- Cephalhematoma: not cross suture line
- *Anterior fontanel and Posterior fontanel*

Cephalhematoma



Caput Succadaneum



- **Craniosynostosis:** premature closure of one or more cranial suture.

Types:

- **Scaphocephaly:** sagittal synostosis
- **Brachycephaly:** coronal synostosis.
- **acrocephaly** :coronal, sagittal, and lambdoid
- **plagiocephaly** :single suture on one side of head

- **Eye:** icteric sclera, conjunctival discharge, paler, erythematic, epicathial folding, sun set eye,
- **Ear:** form, shape, recoil, patency, low set ear, discharge,
- **Nose:** patency, flat nasal bridge; polyp, nasal flaring

P/E FOR NEONATE..

Face: Milia (white cysts) disappear within 2 months without treatment

Throat/mouth: tongue size, arched pallet, cleft lip/pallet, micrognathia, cyanosis ,

- uvula midline, gag reflex,
- pink and moist gum, able to swallow.

Neck: short neck, skin fold, Trachea on midline, Good range-of-motion

P/E FOR NEONATE..

5.Lymphoglandular system:

- breast (Nipples prominent ,Milky secretion (witch's milk)),
- lymph node size

6. Respiratory/ chest:

- Distress signs(Grunting,Tachypnea,Nasal flaring,retraction),
- Deformities, Auscultate ,
- hyper inflation, short sternum ,
- chest circumference(30-33cm) .

7. Cardiovascular system-

- Take apical heart rate for 1 full minute,
- Palpate pulses,
- Assess for cyanosis,
- Pericardium (Active, silent),
- Apical impulse , PMI ,
- Cardiomegaly ,
- Murmur,
- Observe for cardiac distress when feeding.

P/E FOR NEONATE..

8. Abdomen-

- Full Move with respiration , shape
- Distention, Abdominal wall defect
- Kidneys are normally palpable,
- Liver 2-3 cm, Spleen may be palpable,
- Umbilical vessels(2 artery, one vein),
- Hernias ; umbilical and inguinal ,
- bowel sound, fluid, perforated anus,
- Cord(bleeding or drainage , secured).
- physiological jaundice appears

9. Genitourinary system:

➤ FEMALE

- Labia edematous, clitoris enlarged,
- Smegma presents (thick, white mucus discharge from genitalia),
- Pseudo menstruation may present
- Hymen tag may be visible,
- First voiding should occur within 24 hours).
- Presence of vagina.

- **MALE:**
- Prepuce (foreskin) covers glans penis,
- Scrotum edematous,
- Meatus at tip of penis,
- Testes descended but may retract on cold,
- Assess for hernia or hydrocele ,
- First voiding should occur within 24 hours
- Hypospadiasis ,Epispadiasis
- Genital size, Ambiguous genitalia.

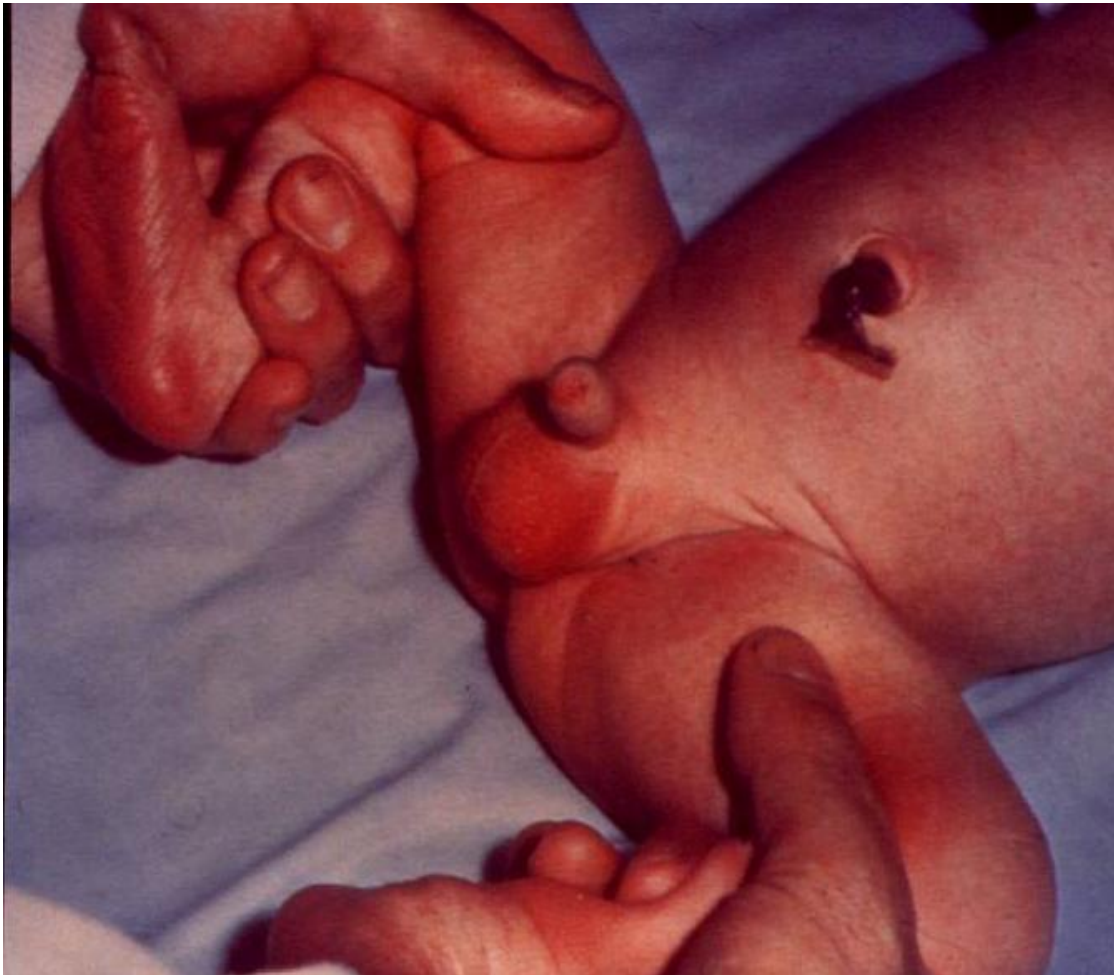
Inguinal Hernias



10. Musculoskeletal system(muscle/bone):

- Flexed, Full ROM, movements symmetrical,
- Fists clenched, syndactyl/Polydactyl
- fractures (clavicle or dislocations (hip),
- feet deformity(club foot)...

Subluxation of the Hip



Common feet abnormalities

- **Club Feet**



polydactyl



11. Integumentary: Skin

- Pinkish, Vernix caseosa , Lanugo(fine hair) ,
- Milia on forehead or skin,
- Acro cyanosis is normal,
- Jaundice ,
- Common skin rash(Erythema Toxicum, mongolian spot, Benign Pustular Melanosis).

Acrocyanosis



Erythema Toxicum:

- Erythematous macules and
- firm 1-3 mm yellow /white papules or pustules,
- Etiology obscure,
- Pustules contain eosinophils and are sterile,
- Appear in the first 3-4 days of life,
- Range: Birth to 14 days,
- Benign and self limited.

Erythema Toxicum



Mongolian Spots:

- gray to blue black lesions,
- Usually over lumbosacral area and buttocks,
- Accumulation of melanocytes within the dermis,
- Generally fade by age 7 years .

Mongolian spots



Impetigo Neonatorum

- Vesicular, pustular, or bullous lesions
- developing in 2-3 days to 2 weeks of life
- Lesions occur in moist or opposing surfaces of skin
- Treat with antibiotics

Impetigo Neonatorum



Pustular Melanosis of the Newborn

- unknown cause ,benign & self limited



12. Central nerves system:

- **Mental status:** Alert ,Irritable , Lethargy ,Coma ,
Neurologic deficit
- Dysraphism or neural tube defect.
- - jitteriness, marked tremors, and seizures
- **Check Neonatal reflexes**

DYSRAPHYSM(NTD)



Lumbar hair tuft



Neonatal reflexes

- ❖ Also known as developmental, primary, or primitive reflexes.
- ❖ They consist of autonomic behaviors that;
- ❖ do not require higher level brain functioning.
- ❖ They can provide information about **lower motor neurons and muscle tone**.
- ❖ They are often protective and
- ❖ disappear as higher level motor functions emerge.

Suckling

- ❖ Onset: ~**28weeks GA**
- ❖ Well-established: **32-34 weeks GA**
- ❖ Disappears: around **12 months**
- ❖ Elicited by the examiner **stroking the lips** of the infant; the **infant's mouth opens** and the examiner introduces their gloved finger and sucking starts.

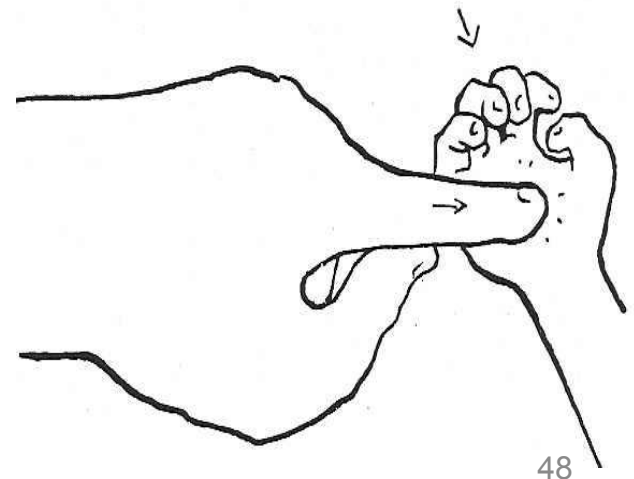
Rooting

- ❖ Onset: **28 weeks GA**
- ❖ Well-established: **32-34 weeks GA**
- ❖ Disappears: **3-4 months**
- ❖ Elicited by the examiner **stroking the cheek or corner of the infant's mouth**. The infant's head turns toward the stimulus and opens its mouth.



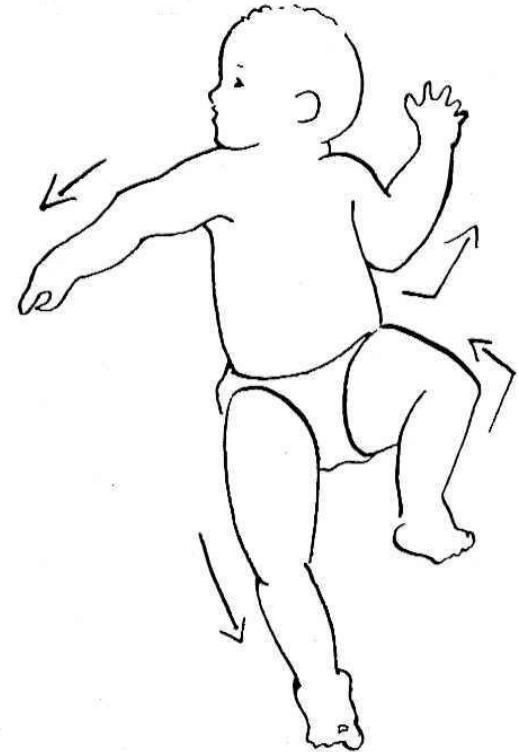
Palmar grasp

- ❖ Onset: **28 weeks GA**
- ❖ Well-established: **32 weeks GA**
- ❖ Disappears: **2 months**
- ❖ Elicited by the examiner **placing his finger on the palmar surface** of the infant's hand and **the infant's hand grasps the finger**. Attempts to remove the finger result in the infant tightening the grasp.



Tonic neck (Fencing posture)

- ❖ Onset: **35 weeks GA**
- ❖ Disappearance: **7 months**
- ❖ Elicited by **rotating the infants head from midline to one side.**
- ❖ The infant should **respond by extending the arm on the side to which the head is turned and flexing the opposite arm.**
- ❖ The lower extremities respond similarly.



Moro

- ❖ **Onset: 28-32 weeks GA**
- ❖ **Well-established: 37 weeks GA**
- ❖ **Disappearance: 6 months**
- ❖ **The examiner holds the infant so that one hand supports the head and the other supports the buttocks.**
- ❖ **The reflex is elicited by the sudden dropping of the head in her hand.**
- ❖ **An absent or inadequate Moro response on one side : hemiplegia, brachial plexus palsy, or a fractured clavicle**
- ❖ **Persistence > 6 months : severe neurological defects.**

Moro

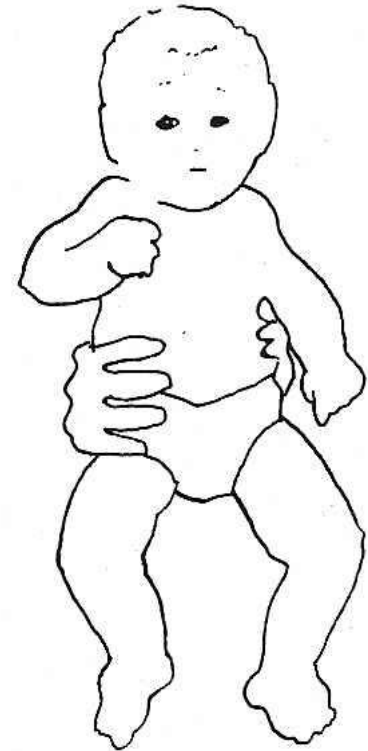
The moro
reflex



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Stepping

- ❖ Onset: **35-36 weeks GA**
- ❖ Well-established: **37 weeks GA**
- ❖ Disappearance: **3-4 months**
- ❖ Elicited by **touching the top of the infant's foot to the edge of a table** while the infant is held upright.
- ❖ The infant **makes movements** that resemble stepping.



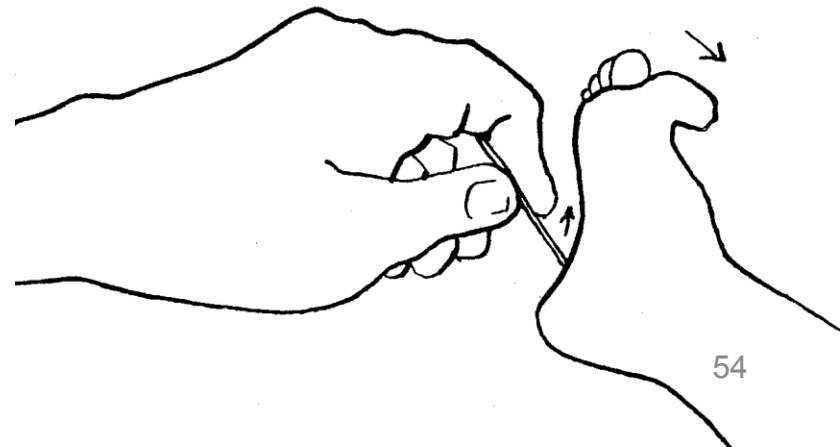
Galant (Trunk incurvation)

- ❖ Onset: 28 weeks GA
- ❖ Well-established: 40 weeks GA
- ❖ Disappearance: 3-4 months
- ❖ The infant is held in ventral suspension with the chest in the palm of the examiner's hand.
- ❖ Firm pressure is applied to the infant's side parallel to the spine in the thoracic area.
- ❖ The response consists of flexion of the pelvis toward the side of the stimulus.



Babinski

- ❖ Onset: **34-36 weeks GA**
- ❖ Well-established: **38 weeks**
- ❖ Disappearance: **12 months**
- ❖ Elicited by stimulus applied to the outer edge of the sole of the foot.
- ❖ The infant responds by plantar flexion and either flexion or extension of the toes.



Classification of newborn

1. *maturity / based on GA only*

- ❖ A. extremely preterm: before 28 WK of GA.
- ❖ B. very preterm: 28-32WK
- ❖ C. preterm :32-37 WK
- ❖ D. term: 37-42WK
- ❖ E . Post term >42WK

2. Birth weight

- ❖ A. **extremely** very low birth weight: < 1000gm
- ❖ B. **very low** birth weight: 1000 - 1500gm
- ❖ C. **low** birth weight: 1500 -2500 gm
- ❖ D .**normal birth wt: 2500-3999gm**
- ❖ E . **Big** baby: >4000 gm

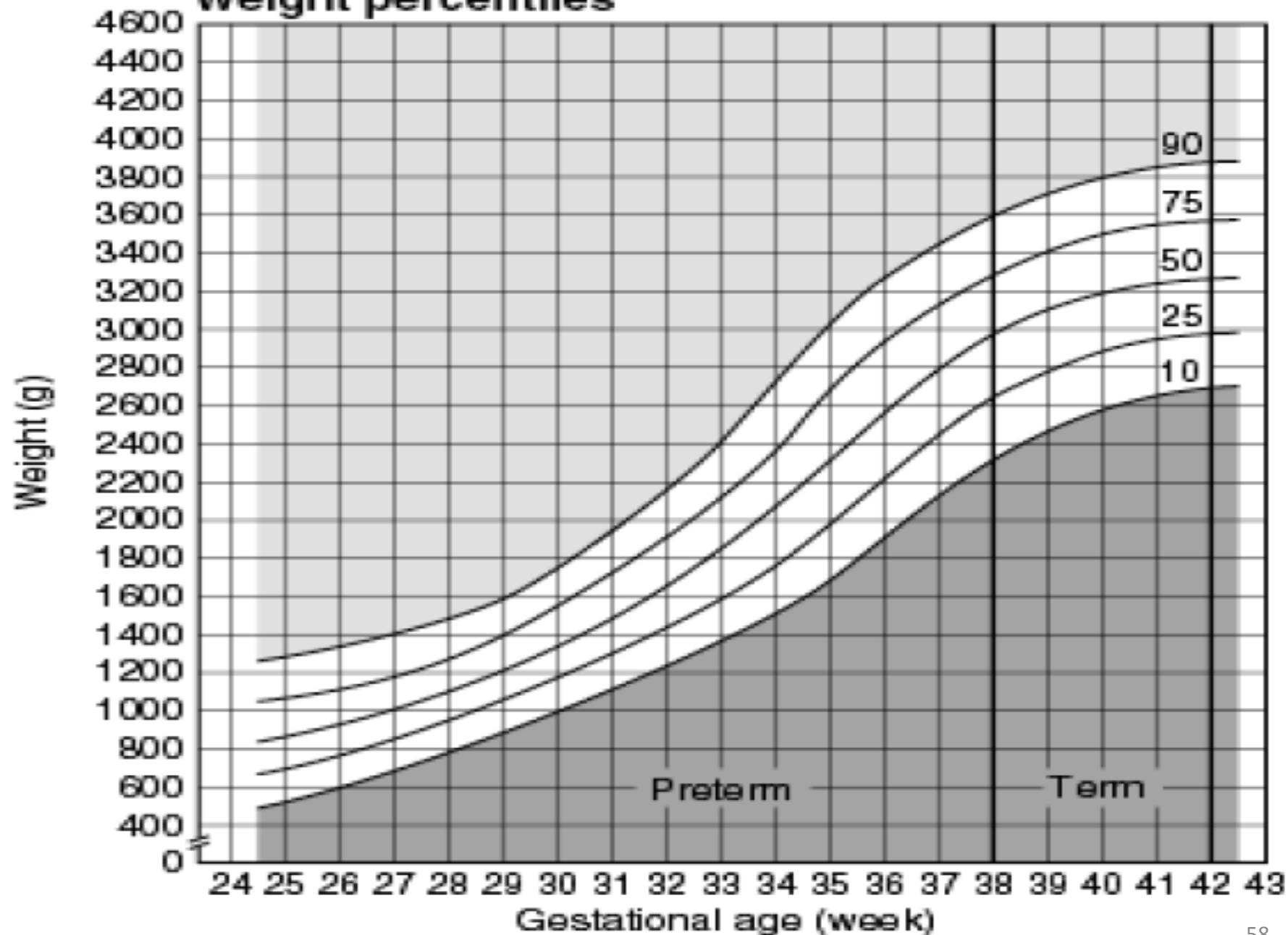
3. Combining the birth wt and GA

A. AGA : birth b/n the 10th and 90th

B. SGA : less than 10th

C. LGA : greater than 90th

Weight percentiles



Newborn danger signs

- Refer baby urgently
- *Unable to feed or sucking poorly*
- *Repeated Vomiting*
- *Convulsions*
- *Movement only when stimulated or no movement, even when stimulated*
- *Gasping or breathing < 30 per minute*
- *Cyanosis (Blue tongue & lips)*
- *Fast breathing (>60/minute, counted 2 times) or severe chest indrawing*

Newborn danger signs....

- *Fever (hot to touch or axillary temp $\geq 37.5^{\circ}\text{C}$)*
- *Hypothermia (cold to touch/ axillary T $<35.5^{\circ}\text{C}$)*
- *Severe jaundice (observed at <24 hrs or ≥ 14 days of age, or involving soles & palms)*
- *Pallor or bleeding from any site*
- *Red swollen eyelids and pus discharge from the eyes*
- *Very small baby (<1500 gm or <32 weeks GA)*
- *Any other serious newborn problem*

Essential newborn care (ENC)

1. Dry and Cleanliness
2. Check breathing/APGAR score
3. Cut cord
4. Skin to skin contact(Thermal protection)
5. Early and exclusive breast-feeding
6. Eye Care(TTC)
7. Chlorohexidine on cord
8. Vitamin k
9. Record (weight ,temp, APGAR score).
10. Immunization(HepB BD ,opv o ,BCG)

Immediate Newborn Care After Birth

Step 1

Dry baby's body with dry and warm towel.
Wipe eye, as you dry stimulate breathing.
Wrap with another dry towel and cover the head while the baby is on mother's abdomen.



Step 2

Assess Breathing, while drying, and manage accordingly - See BIRTH ASPHYXIA chart .

Step 3

Clamp/tie the cord two fingers from abdomen and another clamp/tie two fingers from the 1st one.

Cut the cord between the 1st and 2nd clamp/tie.

Step 4

Place the baby in skin-to-skin contact with the mother.

Keep all babies without complication, in skin to skin contact for one hour.



Step 5

Initiate breastfeeding immediately within 1 hour of life.

Step 6:

Apply Tetracycline eye ointment once on both eyes.



Step 7

Apply Chlorohexidine on the cord.



Step 8

Give Vitamin K, 1 mg IM on anterior mid lateral thigh. (If GA < 34 wks or Wt < 1,500gm , 0.5 mg).

Step 9

Measure baby temperature, Weigh baby & classify - See BIRTH WT & GA Chart.

NOTES

a) If baby needs resuscitation cut the cord immediately.

Otherwise, wait for 1- 3 minutes.

b) Place newborn identification band on the wrist or ankle.

c) Don't forget to record what is done to the newborn.

d) Give HepB-BD, BCG and OPV 0 before discharge.

e) Delay bathing of the baby for 24 hours after birth.

f) Advise mother to apply Chlorohexidine on the cord daily for 7 days and NEVER apply to the eyes.

g) If mother has respiratory infections (like flu, Covid), advise her to wear face mask and to wash her hands while caring the newborn,

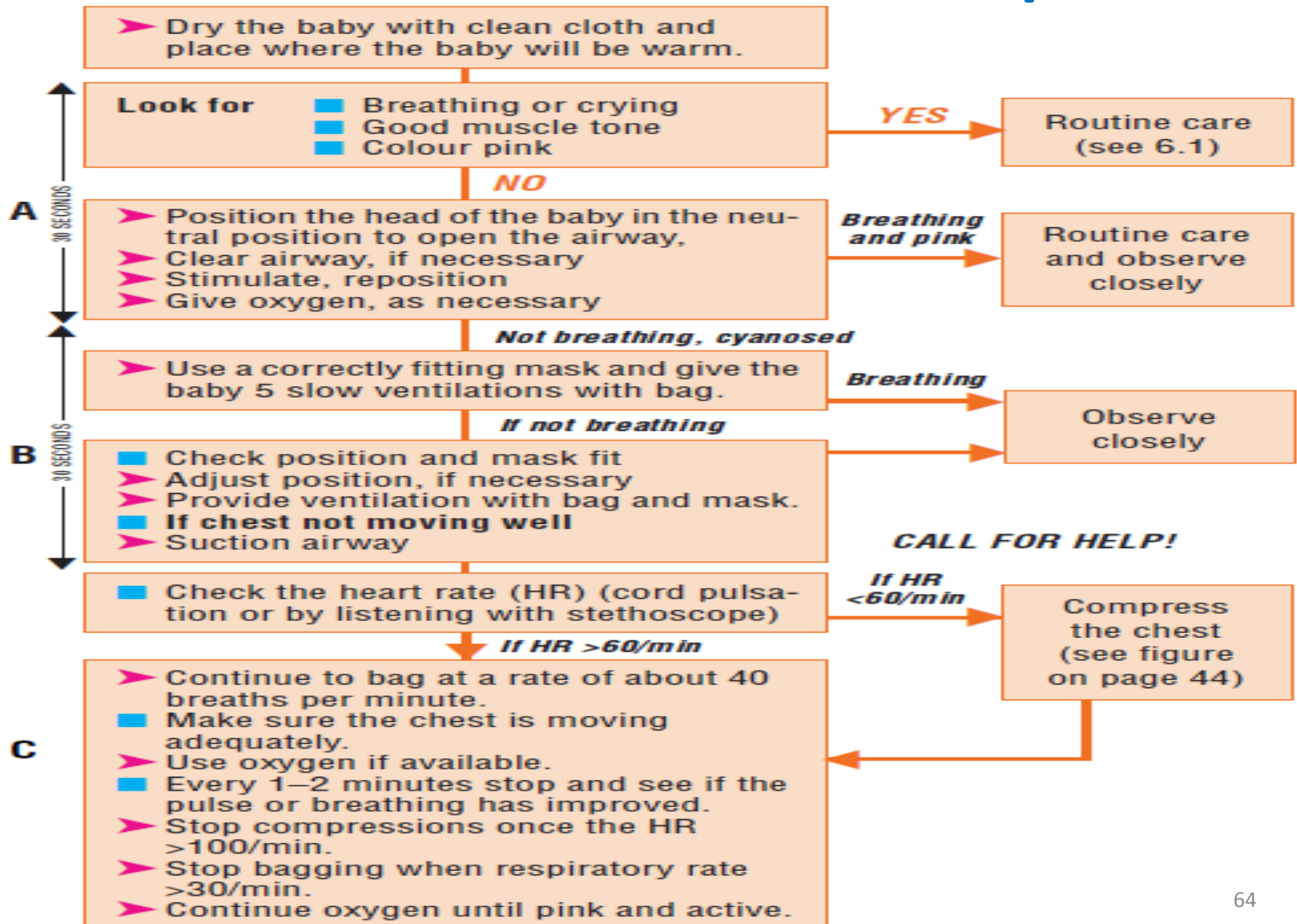
h) Provide postnatal visits at 6 - 24 hours, 3 days, 7 days and immunization visit at 6 weeks.

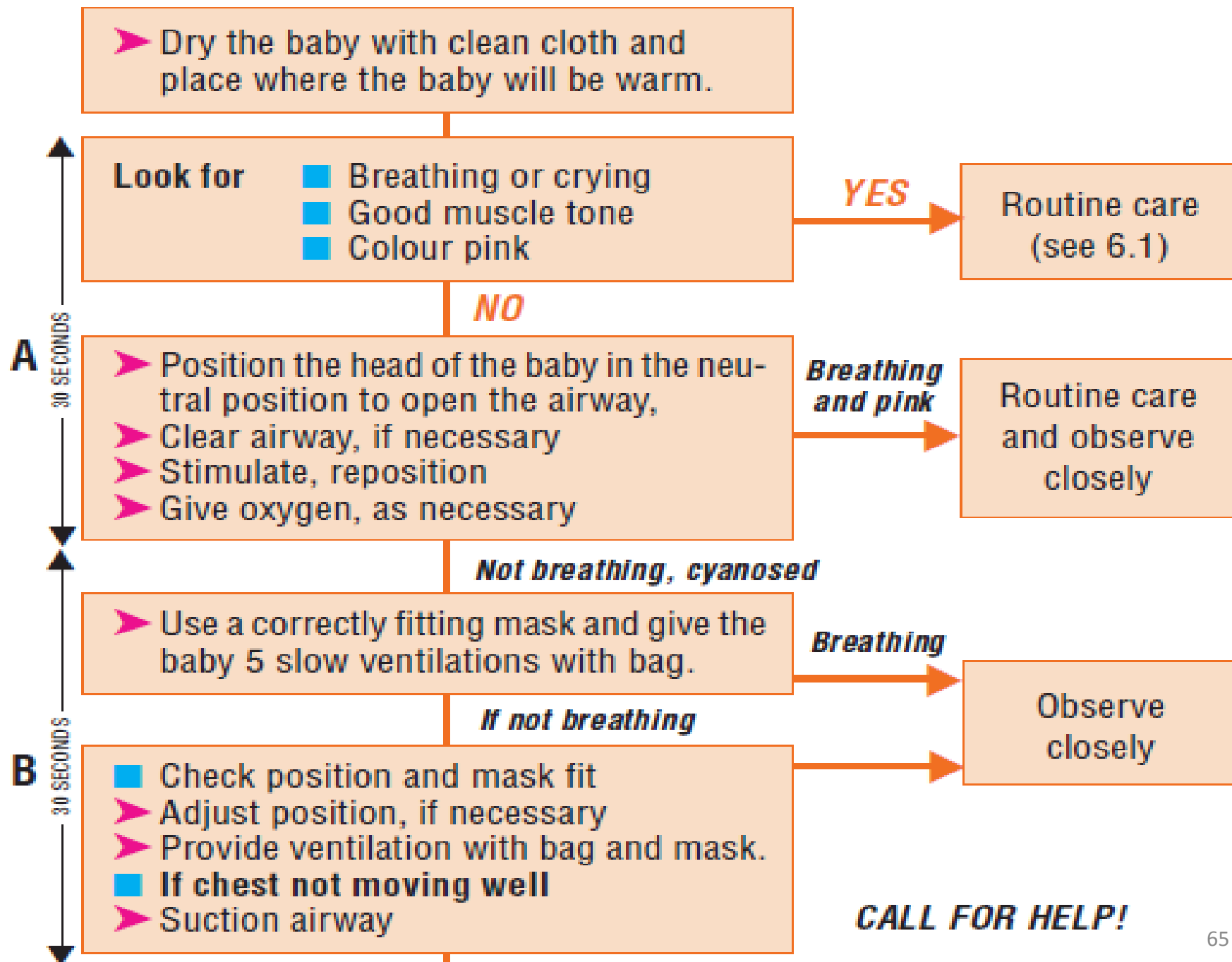
APGAR score

SIGN	0	1	2
Color (<u>A</u>pppearance)	Blue, pale	Body pink, extremities blue	Completely pink
Heart rate(<u>P</u>ulse)	Absent	Below 100	Over 100
Response to catheter in nostril (<u>G</u>rimace)	No response	Grimace	Cough or sneeze
Muscle tone (<u>A</u>ctivity)	Limp	Some flexion of extremities	Active motion
<u>R</u>espiratory effort	Absent	Slow, irregular	Good, crying

- **7-10 well adjusted- dry and Stimulation**
- **4-6 mild asphyxia- suctioning**
- **< 3 severe asphyxia-needs resceustation**

Neonatal resuscitation steps





- Check the heart rate (HR) (cord pulsation or by listening with stethoscope)

*If HR
<60/min*

Compress
the chest
(see figure
on page 44)

If HR >60/min

- Continue to bag at a rate of about 40 breaths per minute.
- Make sure the chest is moving adequately.
- Use oxygen if available.
- Every 1–2 minutes stop and see if the pulse or breathing has improved.
- Stop compressions once the HR >100/min.
- Stop bagging when respiratory rate >30/min.
- Continue oxygen until pink and active.

Newborn resuscitation.....

- Compressions - breathing ratio **3:1**
90 compressions and 30 breaths/minute
 - Stop chest compressions when HR > 100/min
 - Stop ventilation breaths when RR > 30/min
 - If there is no pulse & the baby is not breathing after 20 minutes of resuscitation – stop.
- ❑ **Adrenalín**, if despite art. Ventilation with O₂ and bradycardia <60/min.....???

CPR on neonate



A One over the other



Side by side



B

Fluid preparation

- Prepare 10 % dextrose from;
- 85% ml of D5W and 15% ml of 40% dextrose or 75%ml of NS and 25% ml of 40% dextrose or
- 40% dextrose= Total volume divided by 7, the rest is D5W.
- MF :First day 10% glucose then $\frac{1}{3}$ NS + $\frac{2}{3}$ D10%

Table 4: Fluid requirement of newborn in ml/kg/d

Birth weight in grams.	Day - 1	Day - 2	Day - 3	Day - 4
< 1000	100 - 120	120 - 130	130 - 140	140 - 150
1000-1500	80 - 100	100 - 120	120 - 140	140
1501-2000	80 - 90	90 - 110	110 - 130	130 - 140
>2000	60 - 80	80 - 100	100 - 120	120 - 140

Cont...

- According to **Holiday and Seger formula**, **calorie requirement** is :
- up to 10 kg : 100 kcal/kg
- in the first weeks =120 kcal/kg/day
- After 7 days up to 160kcal/24hrs
- Feed preterm every 2 hours
- Feed term every 3 hours

APPENDIX 10. GUIDELINE TO START ORAL FEEDING

Weight (gm)	INTERVAL	STARTING (ml/kg/d)	INCREMENT (ml/kg/d)	MAXIMUM (ml/kg/d)
<750	Every 2 hrs	10	15	150
750 - 1000	Every 2 hrs	10	15 - 20	150
1001 - 1250	Every 2 hrs	10	20	150
1251 - 1500	Every 3 hrs	30	20	150
1501 - 1800	Every 3 hrs	30	30	150
1801 - 2500	Every 3 hrs	40	40	165
>2500	Every 3 hrs	50	50	180

THE COMMON NEONATAL PROBLEM

- ✓ *Preterm, LBW and SGA infants*
- ✓ *Hypothermia*
- ✓ *Hypoglycemia*
- ✓ *HMD/RDS*
- ✓ *MAS*
- ✓ *PNA*
- ✓ *Neonatal sepsis*
- ✓ *Neonatal jaundice*

PRETERM, LBW AND SGA INFANTS

- **Preterm infants:** <37 weeks gestational age.
- **Low birth weight:** < 2500gram.
- **SGA :** < 10th percentile for gestational age.

Causes of PRETERM, LBW AND SGA INFANTS

- ***Fetal*** - Fetal distress , Multiple gestation , Erythroblastosis ,
- ***Placental*** - Placenta previa , Abruptio placentae.
- ***Maternal*** – Preeclampsia, Chronic medical illness , Infections
- ***Other*** - Premature rupture of membranes , Polyhydramnios

Main problems of preterm newborns

- Hyaline membrane disease , Hypothermia ,
- Hemorrhagic disease of the newborn
- Apnea ,Hyperbilirubinemia, Hypocalcemia ,
- Bacterial sepsis, Hypoglycemia
- Necrotizing enterocolitis , PDA
- Intraventricular hemorrhage ,Anemia , Asphyxia

In addition preterm infants frequently have;

- Weak and uncoordinated ability to feed ,
- Prolonged failure to gain weight and
- Late metabolic acidosis

Main causes of death in low birth weight infants:

- Hyaline membrane disease ,
- Intraventricular hemorrhage ,
- Septicemia ,
- Asphyxia

Problems of SGA or IUGR infants

- Perinatal asphyxia ,
- Hypoglycemia ,
- Hypothermia ,
- Meconium aspiration
- Necrotizing enterocolitis ,
- Polycythemia(HCT > 65 %).

Discussion

**What are the care given for PRETERM, LBW
AND SGA INFANTS?**

❖ Nursery care for PRETERM, LBW AND SGA INFANTS

- *Thermal management*
- *Kangaroo mother care*
- *Initiation of Feeding*
- *Treat the complication*
- *Supplementation with oxygen*
- *Antibiotics (if N. sepsis)*

HYPOTHERMIA

- If Skin Temp < 36.5 and core < 35.5 .
- The newborn is most vulnerable to hypothermia
- especially LBW babies are at risk
- After birth , the skin and core To fall by 0.1 and 0.3/min respectively ;
- Which is equivalent to heat loss of 200kcal/kg body weight/minute.

Mechanisms of heat loss:

- Convection,
- Conduction,
- Radiation,
- Evaporation(common)

Mechanism of heat production:

- muscular activity,
- metabolic thermogenesis

Classification of hypothermia

- 1-Mild Hypothermia(cold stress):** Temp 36-36.4 °C
- 2- Moderate Hypothermia :** 32.0 - 35.9 °C
- 3. Severe Hypothermia :** < 32 °C.

Causes of hypothermia

- External environmental : cold area
- Poor ability to conserve heat in LBW infants
- Poor metabolic heat production
 - Hypoglycemia, Cold delivery room
- Incorrect care of the baby at birth
- Deficiency of fat, e.g. Preterm and SGA
- Delayed initiation of breast-feeding

Consequences of hypothermia :

- Hypoxia,
- Hypoglycemia
- Depressed immunity → sepsis .

Clinical manifestation hypothermia

- Cold feet to touch, Cold skin, Lethargic
- slow, shallow and irregular breathing, Bradycardia
- Low blood sugar and metabolic acidosis,
- Respiratory distress, Acrocyanosis ,
- Poor feeding, hypoxia.

Diagnosis- Based on HX, PE.

Discussion

- **How we manage and prevent hypothermia in neonate ?**

Management of hypothermia

- ✓ re-warmed as soon as possible.
- ✓ There are 3 methods:
 1. Kangaroo mother care,
 2. Warming using a radiant heater and
 3. Warming using incubator.
- Keep infants and room warm,
- 10% dextrose and oxygen
- treatment of infection,
- The temperature should be checked every hour

Prevention of hypothermia:

➤ Warm chain system

- **Components of warm chain system:**
- immediate drying, warm resuscitation,
- skin to skin contact with the mother,
- immediate initiation of breast feeding,
- bathing and weighing post pond,
- appropriate clothing and bedding,
- warm transportation.

HYPOGLYCEMIA

- If glucose **< 40 mg/dL** in **term** neonates or
- If **< 30 mg/dL** in **preterm** neonates.
- Too little glucose can impair the brain's ability to function.
- Severe or prolonged hypoglycemia may result in seizures and serious brain injury.

Risk factors hypoglycemia

- prematurity,
- being small for gestational age,
- maternal diabetes, and
- Placental insufficiency
- perinatal asphyxia.

❖ The most common causes are;

- deficient glycogen stores,
- delayed feeding, and
- hyperinsulinemia.

Etiology...

Causes of transient hypoglycemia :

- Inadequate substrate (eg, glycogen) and
- Immature enzyme function leading to deficient glycogen stores.

Causes of persistent hypoglycemia;

- Hyperinsulinism,
- Defective counter-regulatory hormone release
- Inherited disorders of metabolism

Clinical manifestation of hypoglycemia

- Many infants remain asymptomatic.
- diaphoresis, lethargy or weakness,
- shakiness, jitteriness, seizure, coma,
- cyanotic episodes, apnea, bradycardia or respiratory distress, hypothermia,
- poor feeding, hypotonia, and tachypnea

Diagnosis of hypoglycaemia:

- serum glucose level, serum insulin level,
- urine for ketone.

Treatment of symptomatic hypoglycemia

- Give 2ml/kg of 10% glucose **IV bolus** over one minute
- Give 4 ml/kg of 10% glucoseif seizure is present.
- **10% dextrose for IV bolus** can be prepared using 40% dextrose, which is available in our country by taking one part of 40% dextrose and three parts of distilled water.
- **Example**; for infant weighing 4kgs ,the total volume of 10% dextrose will be $4 \times 2\text{ml} = 8\text{ml}$.To prepare this 8 ml take one part(2ml)from 40% dextrose and three parts (6ml) from distilled water.

Treatment of symptomatic hypoglycemia....

Continuous therapy

- Put on 10% glucose infusion at glucose infusion rate (GIR) of 6mg/kg/minutes (~ 90ml/kg/day) as maintenance.
- Recheck blood glucose after 30 minutes and if it remains above 40mg/dl frequency of checking can be decreased to one hourly then every six hourly.
- If blood glucose remains <40mg/dl, increase the GIR by 2mg/kg/minutes every 30 minutes until repeat values are above 40mg/dl.

Prevention of hypoglycemia

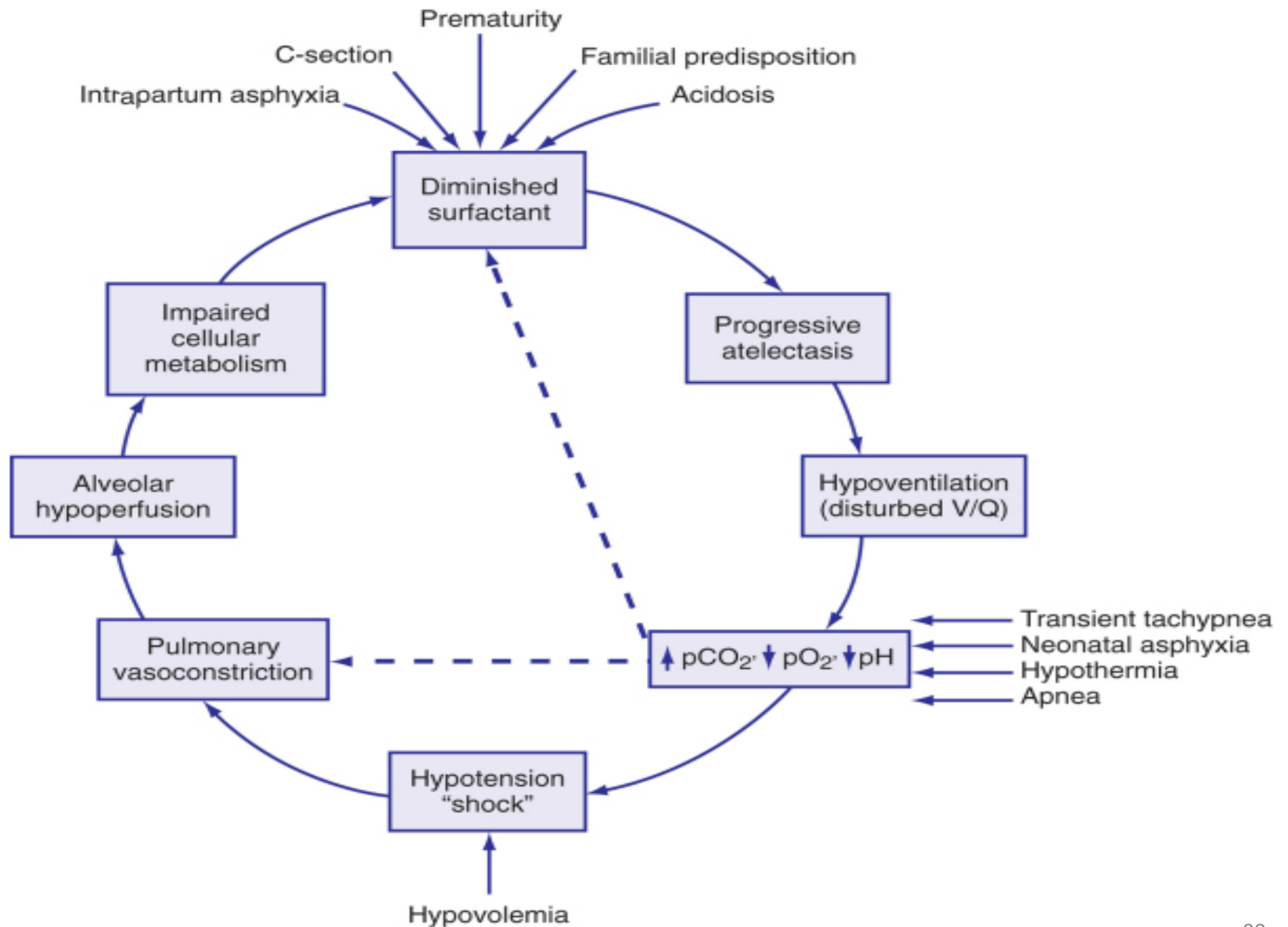
- **Early** feeding of Infants of DMM and SGA.
 - Start 1 hour of age - 3 feed at 1 hour interval, 5ml/kg/feed.
- If normal blood sugar continue feed 2 hrly
- preventing the risk factor(hypothermia, asphyxia.....).

RESPIRATORY DISTRESS SYNDROME (HYALINE MEMBRANE DISEASE)

- premature infants **lack** enough **surfactant** to keep the alveoli open.
- **Hyaline Membrane Disease** : is one of the most important causes of mortality and morbidity in **preterm neonates**
- **Etiology**: **Surfactant deficiency**
- **Respiratory distress**: The presence of any one of the following four clinical features :
 - RR > 60/min, Significant lower chest indrawing,
 - Grunting or Central cyanosis.

Risk factor for HMD

- **Prematurity**
- Maternal DM,
- Male sex,
- 2nd born twins,
- C/S delivery
- Perinatal asphyxia
- cold stress



Clinical features of HMD

- usually appear within minutes of birth,
- Characteristically, **tachypnea, nasal flaring, -subcostal and intercostal retractions, -cyanosis and expiratory grunting** are noted.
- Decreased bilateral air entry
- The natural course is characterized by progressive worsening of cyanosis and dyspnea.

Diagnosis of RDS

- history and physical examination,
- CXR

Treatment of RDS/HMD

- ✓ Thermoregulation
- ✓ Fluid
- ✓ Antibiotics
- ✓ Ventilatory support
- ✓ CPAP
- ✓ Surfactant therapy

Prevention: the prevention of prematurity.

- Administration of dexamethasone to a mother 48 hrs before delivery (born between 24–34 wks)

MECONIUM ASPIRATION SYNDROME.

- Respiratory distress in newborn infants born through meconium stained amniotic fluid.
- The passage of meconium in utero accompanies 8 to 20% of all deliveries and is seen predominantly
- in small for gestational age and post mature infants.
- 10-15% of births-meconium stained amniotic fluid,
- 5% -meconium aspiration pneumonia ,
- 30% require mechanical ventilation and
- 3-5% expire

Risk factors of MAS:

- Placental dysfunction,
- Fetal hypoxia,
- Ante partum haemorrhage,
- Post maturity or SGA ,
- Breech delivery.

Pathophysiology MAS

1. Fetal hypoxia
2. Peristalsis & IU passage of meconium
3. Meconium stained amniotic fluid
4. Meconium aspiration
5. Peripheral & proximal air way obstruction,
6. inflammatory & chemical pneumonitis

Clinical features MAS

- nails, skin and umbilical cord that are heavily stained with a yellowish pigment.
- Respiratory distress with cyanosis, grunting, flaring, retractions and marked tachypnea

Treatment of MAS

- Suctioning
- endotracheal intubation

Prevention of MAS

- Prevent fetal distress
- Careful, thorough suctioning of the oropharynx after the head is delivered.

TRANSIENT TACHYPNEA OF THE NEW BORN

- **Transient tachypnea** is a respiratory disorder seen shortly after delivery
- **Transient** means it is short-lived (< 24 hours).
- **Tachypnea** means rapid breathing (> 60 b/ min).
- occasionally called RDS type II,

TTN...

- It may be characterized by the;
 - early onset of tachypnea,
 - sometimes with retractions, or
 - expiratory grunting and,
 - occasionally, cyanosis that is relieved by minimal oxygen (<40%).
 - Patients usually recover rapidly within 3 days.

Causes of TTN

- **2ndry** to slow absorption of fetal lung fluid
- Transient tachypnea is more likely to occur in babies who were:
 - Born before 38 weeks gestation
 - Delivered by C-section, especially if labor has not already started
 - Born to a mother with diabetes

Clinical manifestation of TTN

- have breathing problems soon after birth,
- Bluish skin color (cyanosis),
- Rapid breathing,
- grunting, Flaring nostrils or
- intercostals retractions.

DX of TTN

- The mother's pregnancy and labor history
- Blood count and blood culture
- Chest x-ray
- Continuous monitoring of v/s
- DX most often made after the baby is monitored for 1 or 2 days.

Treatment of TTN

- oxygen
- Fluids and nutrients will be given through a vein
- Antibiotics until R/O no infection.
- Most infants improve in less than 12 - 24 hours.
- The condition most often goes away within 24 - 48
- They will not need special care

Discussion

- A 33 weeks GA , 2.4 kg male newborn has difficulty of breathing and failure to suckling. His RR is 70b/min
- A 1 hour female neonate presents with heavily stained with a yellowish pigment on her nails, skin and umbilical cord . She has cyanosis, flaring, intercostal retractions. RR is 76 , T-36.7.
- The most likely DX is.....
- Cause of this problem?
- Mgt ?

PERINATAL ASPHYXIA OR HYPOXIA

- it is an **insult to the fetus or newborn** due to;
 - **hypoxia**) and /or **ischemia** to various organs.
- Failure to establish efficient breathing at one minute of age (APGAR score 0 - 6) .
- characterized by **hypoxia and hypercapnia** .
- It is emergency condition and need quick treatment

PERINATAL ASPHYXIA OR HYPOXIA...

Etiology

- **Maternal factor:** hypoxia, anemia, diabetes, hypertension, smoking, heart disease, too old or too young, etc
- **Delivery condition:** Abruption of placenta, placenta previa, premature rupture of membranes, etc
- **Fetal factor:** Multiple pregnancy, congenital or malformed fetus, etc

PERINATAL ASPHYXIA OR HYPOXIA...

Etiology

- 90% due to placental insufficiency.
- Decrease oxygen supply
- Decrease carbon dioxide and hydrogen removal
- <10% are post natal insult due to pulmonary, CVS or neurologic insufficiency

PERINATAL ASPHYXIA OR HYPOXIA...

Clinic manifestations

- Depends on the organ affected.
- Kidney in 50% of cases,
- CNS in 28%,
- CVS in 25% of cases,
- Pulmonary 23% of cases

Clinic manifestations of PNA

- **Central nervous system:** infarction, intracranial hemorrhage, seizure, hypoxic-ischemic encephalopathy
- **Cardiovascular:** tachycardia->bradycardia, ventricular hypertrophy, arrhythmia, hypotension, myocardial ischemia
- **Respiratory system;** apnea, respiratory distress syndrome, cyanosis.

Clinic manifestations of PNA..

- **Renal:** Decrease in UO, Protein & cast may present in the urine ,
 - Gross hematuria, Elevation of BUN& creatinine
- Apgar score of <3 (severe) and 4-6 (moderate),
- Thick meconium-stained amniotic fluid

Hypoxic Ischemic Encephalopathy (HIE)

- The **brain damage after perinatal asphyxia** and
- the most severe condition showed high mortality or remain **cerebral complications** such as **mental retardation** .
- The most and direct cause of HIE is P.asphyxia.
- The neuralgic squale often persists.

Hypoxic Ischemic Encephalopathy (HIE)...

- The clinic features of HIE are mainly symptoms of consciousness;
- **Excitation** :hyperalert, irritable, hypertonia,
-tachycardia, tachypnea, seizure, etc
- **Depression** :coma, hypotonia, bradycardia,
- bradypnea, unresponsiveness, etc

Sarnat staging of hypoxic-ischemic encephalopathy.

	Grade 1 (mild)	Grade2 (moderate)	Grade 3 (severe)
Level of consciousness	Irritable/hyperalert	Lethargy	Coma
Muscle tone	Normal or hypertonia	Hypotonia	Flaccid
Tendon reflexes	Increased	Increased	Depressed or absent
Seizures	Absent	Frequent	Frequent
Complex reflexes	Normal	weak	Absent
Prognosis	Good (100%) Normal	Variable (80%) Normal	High mortality and neurological disability (50% Death 50% major sequelae)

PERINATAL ASPHYXIA OR HYPOXIA...

Investigation:

- **EEG**-severity /presence of seizure,
- **Cranial u/s**- presence ICH & brain edema ,
- **CT scan** –brain edema, encephalomalacia
- **RFT** (BUN, Creatinine)

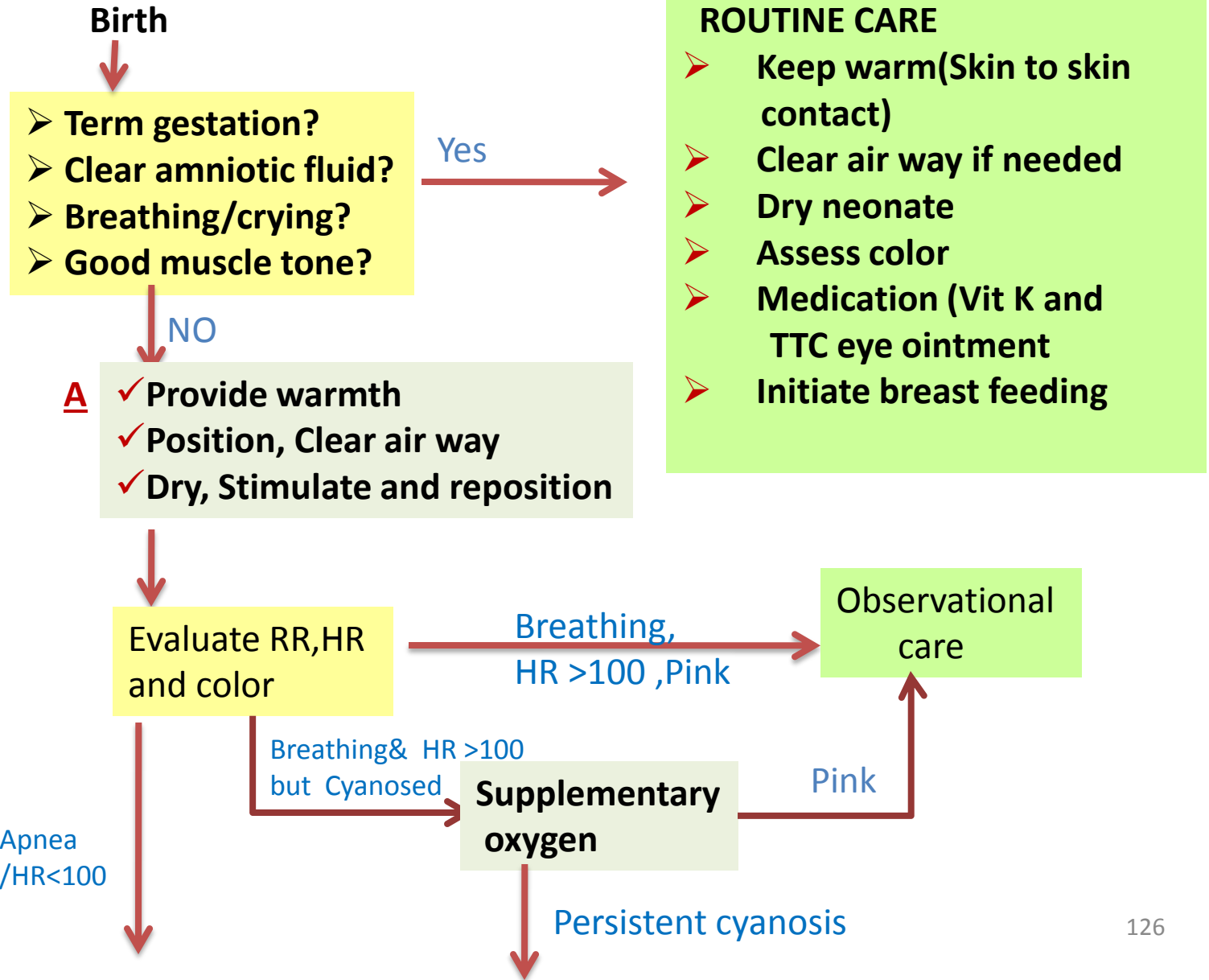
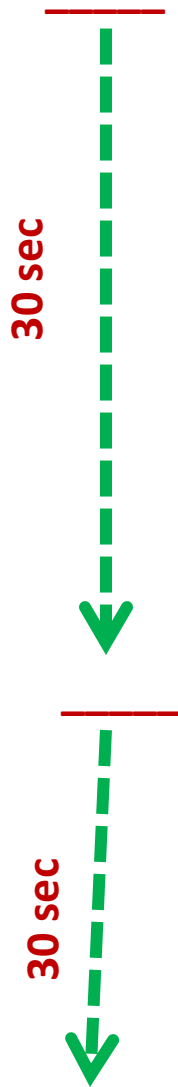
PERINATAL ASPHYXIA OR HYPOXIA...

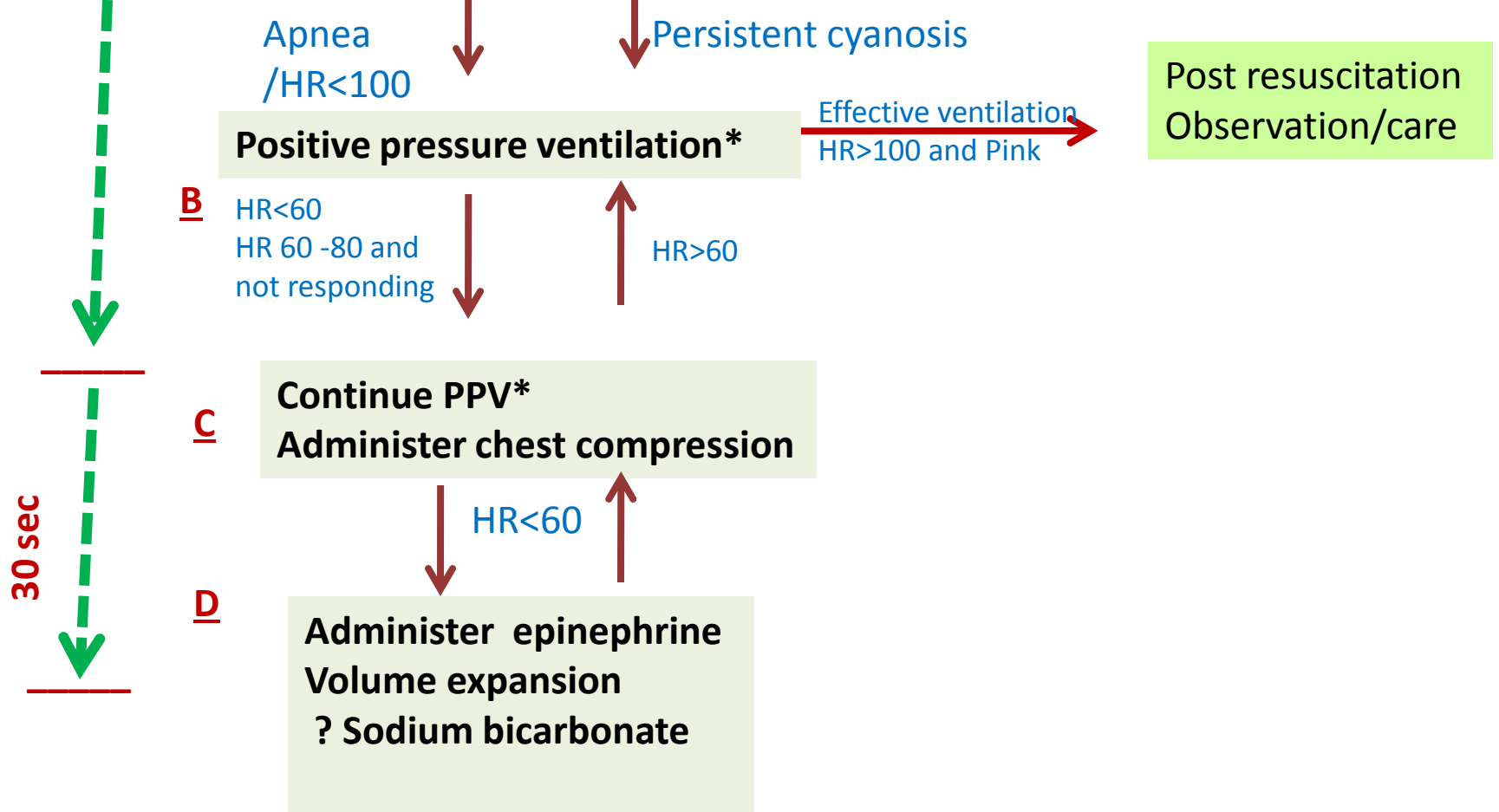
Management of perinatal asphyxia

- Anticipate need for neonatal resuscitation
- **Neonatal resuscitation**
- Avoid blood pressure fluctuation,
- Intravenous fluid 2/3 of maintenance
- Appropriate ventilation support,
- Correct metabolic abnormalities
- Prophylactic anticonvulsant (phenobarbital)..?
- Therapeutic hypothermia

Neonatal Resuscitation flow algorithm

Time frame





Discussion

- You are attending a delivery of a 39 wks' GA. At delivery, you receive a floppy, apneic, and blue term male infant.
- The most likely DX is.....
- Your next step....

NEONATAL INFECTIONS /SEPSIS

- Neonatal sepsis -local or systemic infection in the first month of life.
- It faced to all new borns
- risk and severity↑ small and premature infants.
- It is the cause of 30-50 % of Neonatal mortality
- Sepsis in the Neonate includes ;
- meningitis, septicemias, pneumonia,
- Arthritis, osteomyelitis, and UTI.

Early-onset neonatal sepsis :

- occurs within the first week (<7 days) of life
- Is caused by organisms prevalent in the maternal genital tract, in the labor room or operation theatre
- Majority are caused by ;
- Gp B streptococcus, E-coli, klebsiella and enterobacter
- Majority manifest with respiratory distress

Risk factors for neonatal sepsis

- Birth asphyxia, Unclean vaginal examination,
- Foul smelling liquor,
- Prolonged labor (>24 hrs),
- Preterm and low birth weight neonate,
- Prolonged rupture of membrane (>24 hrs)
- Maternal pyrexia.
- Other congenital immune defects.

Late onset neonatal sepsis :

- occurs after the first week of life (**>7 days**).
- Most are caused by gram negative bacterias;
- Klebsiella, enterobacter, E-coli,
- pseudomonas and salmonella.
- Gram positives like staph aureus and staph albus
- Usual sources include Incubators, resuscitation equipments, feeding bottles catheters, infusion sets and sites.

Clinical features of neonatal sepsis

- Early change in behavior or feeding patterns.
- Lethargic, inactive and unresponsive,
- Failure to suck, Cyanosis,
- hyper/hypothermia,
- Evidences of respiratory problems
- hypotension,
- Failure to gain weight

- **Bacterial meningitis:** About a third of babies with neonatal sepsis can have coexisting meningitis.
- more common with late onset neonatal sepsis.
 - Convulsions/twitching, Staring look,
 - Bulged anterior fontanel
 - Abnormal high pitched cry/excessive cry
 - Abnormal CSF analysis
- **Pneumonia :** common in early onset NS.

Diagnostic work up of NS

- Clinical + presence of high risk obstetric factors,
- Blood culture and sensitivity,
- CSF analysis when indicated,
- Chest X-ray,
- Urin analysis and /or urine culture

Management of neonatal sepsis:

- Depending on the etiologic agent.
- Till etiologic agent is identified:broad spectrum
- **First line drugs:** Ampicillin and Gentamicin
- **Second line drugs:**Ceftriaxone + Aminoglycoside
- In Meningitis, the dose doubled
- Supportive therapy (O2, Fluid.....)

Prevention of neonatal sepsis

- Hand washing before entering labor ward, before and after examining each infant
 - Single most important one .
- Wear gowns and slippers when entering NICUs
- HCP with acute infections (Fever,ARI,GE,Skin lesions and viral exanthems) should be restricted from providing care to the neonates
- Keep clean illness of the neonatal and labor ward(cleaning and fumigating at regular intervals)
- Proper skin and cord care, Keep all equipments clean

CASE STUDY

- A newborn whose estimated gestational age is 38 weeks has had worsening respiratory distress since birth. Maternal history is significant for prolonged premature rupture of membranes & foul-smelling amniotic fluid.
- DX?
- TX?

NEONATAL JAUNDICE

Neonatal jaundice - it is visible jaundice due to elevated serum level of **bilirubin**.

- Occurs in 60% of term & 80% of preterm neonate.
- Is common problem and is mostly benign.
- However, significant jaundice occurs in 6 % of term babies

Hyperbilirubinemia - It is a markedly elevated serum bilirubin usually **>15 mg/dl** which predispose neonates to kernicterus.

unconjugated (indirect reacting)hyperbilirubinemia.

- The most common type
- there is increased production,
- decreased hepatic conjugation, or
- decreased hepatic uptake of bilirubin.
- crosses the BBB bilirubin encephalopathy .
- It is neurotoxic

Conjugated hyperbilirubinemia

- less common in neonates and
- most often denotes potentially serious disorder.
- However, it is not neurotoxic.
- The conjugated bilirubin will go to bowel with bile.
- Conjugation and enterohepatic resorption.

Physiologic Jaundice :

1. result of increased bilirubin production
2. Jaundice becomes visible on the **second or third day**,
 - usually peaking b/n **2-4 days at 5 to 6 mg/dl**, and
 - decreasing between the 5-7 day of life.

Jaundice becomes evident as physiologic B/c :

- Short life span of RBCs(70-90days),
- RBC mass is increased
- Immature ligandine ,
- High activity β -glucuronidase (gut),
- Decreased flora in the gut

Pathological Jaundice

❖ *Pathological Jaundice* should be considered if:-

- It appears in the 1st 24 hours of life
- Serum bilirubin is rising at rate $> 5 \text{ mg/dl/24 hr.}$
- Serum bilirubin is $> 12 \text{ mg/dl}$ in full term or
 - $> 14 \text{ mg/dl}$ preterm infants
- Jaundice persists after >14 days for preterm
 ≥ 8 days for term
- Direct reacting bilirubin is $> 2 \text{ mg/dl}$

Table 15: Features differentiating Physiological jaundice from Pathological Jaundice

<i>No</i>	<i>Features</i>	<i>Physiologic Jaundice</i>	<i>Pathological Jaundice</i>
1	Clinical onset of jaundice (after birth)	>24 hrs	<24 hrs
2	Jaundice still clinically visible (day after birth)	Term < 8 days Preterm < 14 days	Term $\geq$ 8 days Preterm > 14 days
3	Peak Total Serum Bilirubin (TSB)	Term < 12 mg/dl Preterm < 15 mg/dl	Term > 12 mg/dl Preterm > 15 mg/dl
4	Rise in TSB	< 5mg/dl/24 hrs	> 5mg/dl/24 hrs
5	Conjugated serum bilirubin level	<2mg/dl	>2mg/dl or 15 % of TB

Etiology of jaundice

Causes of unconjugated hyperbilirubinemia

- Increased load of bilirubin eg. hemolytic anemia
- Damage or reduction of the activity of transferase enzyme (genetic deficiency , hypoxia, prematurity, and infection) ,
- Blockage of the transferase enzyme

Etiology of jaundice...

Causes of conjugated hyperbilirubinemia

- Sepsis ,
- Intrauterine infection,
- Biliary atresia ,
- Hepatitis

Etiology of jaundice...

Cause: Jaundice appearing in the 1st 24 hr:

- Rh incompatibility(erythroblastosis fetalis),
- ABO incompatibility,
- Defect (G6PD def. and spherocytosis)
- TORCHS,
- Criggler Najar syndrome,
- transient familial neonatal jaundice.

Etiology of jaundice...

Jaundice appearing with in 24-72hr of age

- Physiological jaundice
- Cephalhematoma ,
- Hypothermia
- Prematurity,
- Hypoglycemia,
- Multiple bruises,
- hypoxia

Etiology of jaundice...

prolonged jaundice(after 72 hrs.):

- TORCHS,
- Sepsis,
- Hypothyroidism,
- Biliary Atresia
- Breast milk jaundice,
- Drugs (vit. K, oxytocin, diazepam),
- Metabolic disease eg. galactosemia

Etiology of jaundice...

Jaundice associated with breast feeding:

Breast milk:

- cause is ?glucouronidase in breast milk,
- time of onset after first week and
- treated by discontinuation of BM for 2 – 3 days.

Breast feeding:

- cause is reduced intake ,
- time of onset within one week and
- treated by continuing breast feeding.

Clinical Manifestations neonatal jaundice

- Jaundice
- Jaundice begins on the face-abdomen -feet.
- lethargic or may feed poorly.
- Pallor, petechia , Bruises and cephalhematoma,
- Dark urine and clay colored stool.
- Jaundice on face- 5 mg/dL;
- mid-abdomen-15 mg/dL; soles-20 mg/dL),

Kernicterus

- **Kernicterus** is a neurologic syndrome resulting
- from the deposition of unconjugated bilirubin in the brain cells.
- High Risk ;Preterms and newborns with infection

Kernicterus Clinical manifestation :

- lethargy, poor feeding and loss of moro reflex.
- diminished deep tendon reflexes and
- respiratory distress.
- opisthotonus, bulging fontanel,
- a high pitched cry and convulsions
- Overt neurologic signs have a grave prognosis:
 - 75% or more of such infants die
 - 80% of affected survivors have
 - Bilateral choreoathetosis with involuntary muscle spasms
 - Mental retardation, Deafness and Spastic quadriparesis

Management of neonatal jaundice

- **Aim** : lower serum bilirubin and
 - decrease neurotoxicity

Principles of treatment

- Avoid drugs w/c interfere with BBN metabolism,
- Treat factors w/c ↑ neurotoxicity
- Give adequate feeding, Specific therapy,
- Decrease serum bilirubin
- **Lower serum Bilirubin by**
 - **Phototherapy** (cover eye & genital, change every 2 hrs from supine to prone , 25% additional fluid)
 - **Exchange transfusion(most effective)**

NECROTIZING ENTEROCOLITIS (NEC)

- where portions of the **bowel** undergo **necrosis**
- typically occurs in second to third week of life in premature, formula-fed infants.
- primarily seen in premature infants,
- it may occur up to 3 months after birth.

Cause of NEC

- The exact cause of this disorder is unknown.
- A drop in blood flow
- Bacteria in the intestine
- premature infants have an undeveloped immune response to factors
- An imbalance in immune regulation appears

Clinical feature: of NEC

- Blood in the stool ,Vomiting ,abdominal distension
- Diarrhea, Feeding problems, Lack of energy,
- Unstable body temperature,
- Unstable breathing, heart rate, or blood pressure,
- intestinal perforation and peritonitis and
- systemic hypotension, hematuria,
- abdominal mass, Erythema of the abdominal wall,
- Systemic signs :Respiratory failure, Decreased peripheral perfusion, Circulatory collapse

Diagnosis of NEC

- clinically, CBC ,serum electrolyte ,
- abdominal X-ray and Ultrasound.
- Plain abdominal radiographs are essential to make a diagnosis of NEC. The finding of pneumatosis intestinalis (air in the bowel wall) confirms
- Stage I: Mild symptoms
- Stage II: moderate symptoms(metabolic.....)
- Stage III: Shock , perforation

Treatment of NEC

- Stop enteral feedings
- Perform nasogastric decompression
- Initiate broad-spectrum antibiotics
- Keep NPO
- IV fluids, including total parenteral nutrition
- *Surgery*

Neonatal tetanus

- acute, **spastic paralytic illness** caused by tetanus toxin, the **neurotoxin** produced by;
- ***Clostridium tetani***, a motile, gram-positive, spore-forming anaerobe bacteria.
- **C. tetani** is found every where
- Transmitted via unhygienic delivery
- Maternal immunization(TT) protects the baby
- An incidence of neonatal tetanus **<1/1000** live birth is a cutoff point for **eradication**

Neonatal tetanus...

PATHOGENESIS

- Acquired via cord contamination
- Incubation period (3-14 days)
- Spore germinate up on entry in to the tissue
- Vegetative cells release tetanospasmin
- The toxin bind nerves and ascend to the spinal cord inhibiting motor neurons
- Toxin inhibit relaxation of muscles by inhibiting GABA
- No effect on the brain tissue

tetanus....

Clinical manifestations

- First sign is **failure to suck** between 3-14 days of life due to oral muscle spasm (lock jaw)
- **Stiff neck**, difficulty of swallowing
- Weak cry to gurgling sound
- **Body arching** due to generalized muscle spasm (**Opisthotonus**)
- **sardonic smile of tetanus (risus sardonicus)** results from intractable spasm of facial and buccal muscles.

Neonatal Tetanus

Suspected case

- **Neonatal death** who has been **feeding and crying** normally for the **first two days**

Confirmed case (three of them)

- Neonate healthy for the first two days
- Illness between the age of 3 to 28 days
- **Failure to suck (trismus or lock jaw)**
- **Stiffness (rigidity) and convulsions (seizure)**

Neonatal tetanus

Diagnosis

- Clinical presentation
- Laboratory confirmation is difficult

Differential diagnosis

- Asphyxia
- Meningitis
- Seizure disorder
- Kernicterus
- Hypocalcaemia tetani

Neonatal tetanus

Complications

- Aspiration pneumonia
- Laryngo-spasm—asphyxia—coma--- death
- Hypertension,
- hypotension,
- arrhythmia
- Seizure
- frequent spasms
- Spasm related bone fracture

Neonatal tetanus

Management

1. **Control spasms**: Diazepam, silent room
2. **Control toxin**: TAT, human antitoxin
3. **Eradicate C. tetani**: Metronidazole or Crystalline penicillin
4. **Wound care**: debridement and cleaning
5. **Supportive care**: Oxygen, IV fluids, NGT feeding, correct electrolytes
6. **Tracheostomy**: for air way obstruction

Prevention

- Routine immunization of mothers with TT
- Hygienic delivery and cord care

HEMORRHAGIC DISEASES OF THE NEW BORN

- also known as **Vitamin K Deficiency Bleeding**
- As a consequence of vitamin K deficiency there is
- an impaired production of coagulation factors;
- II, VII, IX, X, protein C and protein S by the liver.
- It **most** often develops **shortly after a baby is born**.

VITAMIN K type:

- **vitamin K1**- occurs in some plants and
vitamin **K2**- synthesized by bacteria(E-Coli).
Vitamin **K3** - synthetic substance

Cause of HDN

- Vitamin K does not move easily across the placenta from the mother to the baby.
- vitamin K in breast milk is low
- gut flora has not yet been developed
- Breast feed infants have up to 20 times greater risk of developing HDN

Cause of HDN

- vitamin K malabsorption;
 - neonatal hepatitis or biliary atresia
- Deficient hepatic stores
- pancreatic disease, and intestinal disorder
- Prolonged diarrhea

HDN....

1. Early onset ;occurs first hours -24 hrs, very rare.

- Use of anti-seizure drugs or a blood thinner

2. Classic onset : between 24 hours and 7 days after birth.

- in breast-fed infants who did not receive a vitamin K shot
- It is also rare.

3. Late onset: b/n 2 weeks and 2 months old.

- It is more **common** in children who did **not receive a vitamin K** shot.

Clinical presentation of HDN

- The disease causes an increased risk of **bleeding**.
- The most common sites of bleeding are
 - **umbilicus, mucous membranes,**
 - **gastrointestinal tract,**
 - **circumcision and**
 - **venepunctures.**
- Others bleeding in urine, bruising , bleeding under skull bone.....

Diagnosis of HDN

- Proper history and physical examination,
 - Blood clotting tests will be done.
 - A PT should be measured in cord blood
 - CT scan to evaluate intracranial bleeding
- The diagnosis is confirmed if ;
- a vitamin K shot stops the bleeding and
 - blood clotting time (prothrombin time) is within normal limits.

Treatment of HDN:

- vitamin K supplementation.
- transfusion of fresh frozen plasma or
- whole blood

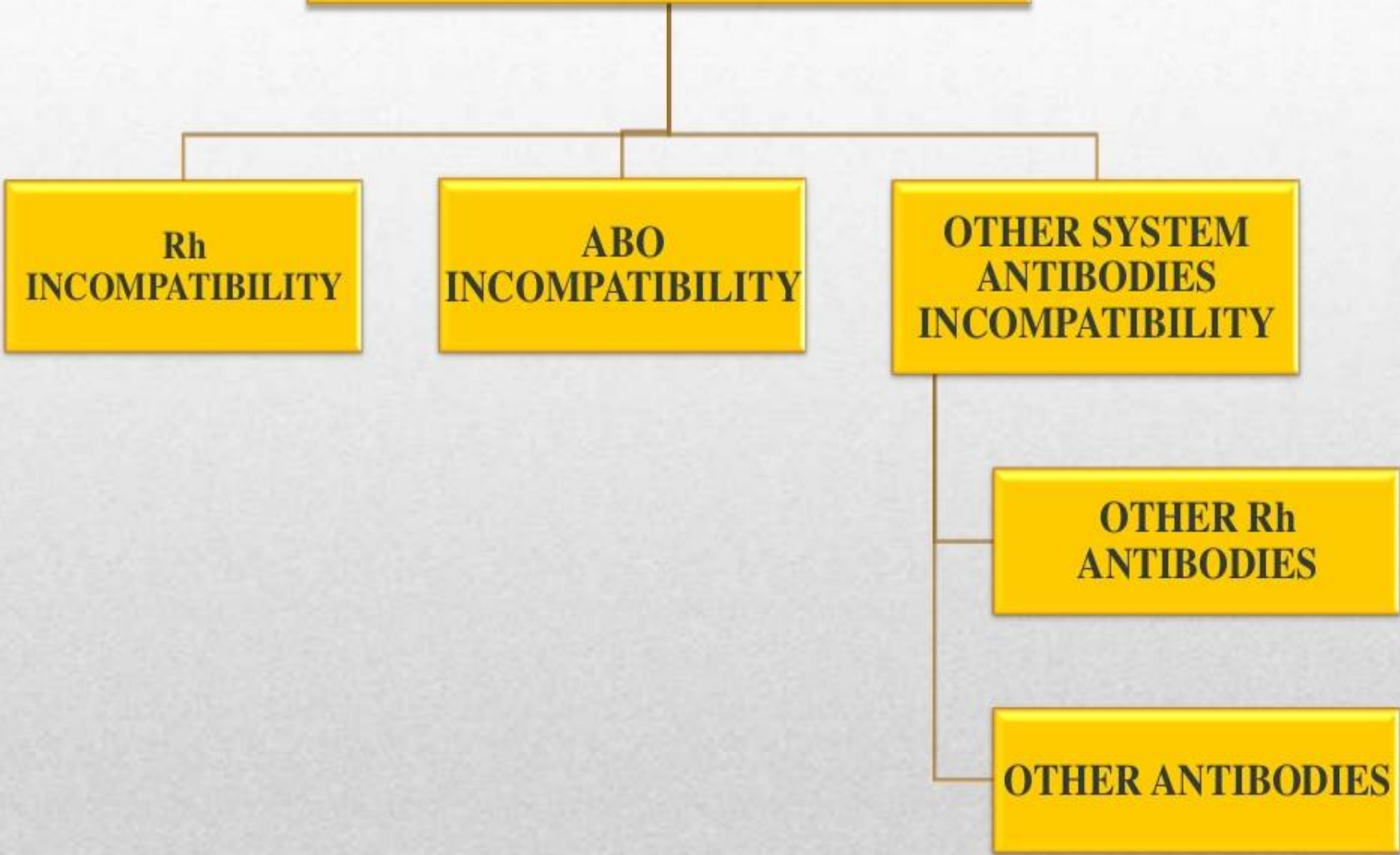
HEMOLYSTIC DISEASE OF NEWBORN (ERYTHROBLASTOSIS FETALIS)

- caused by the transplacental passage of **maternal antibody** active against **paternal RBC antigens of the infant** .
- It occurs due to **destruction of fetal red cells**;
- by **IgG antibodies** from maternal blood against red cell antigens acquired from the father.

HEMOLYSTIC DISEASE OF ..

- Needs an initiation *factor* (***mixing of blood*** even in small amount)
- so sensitization **rarely occur in first pregnancy.**
- increased rate of RBC destruction,
- short life span of the fetal/neonatal **RBC**
- It is an important cause of anemia and jaundice in newborn infants

CAUSES OF HDN



```
graph TD; A[CAUSES OF HDN] --> B[Rh INCOMPATIBILITY]; A --> C[ABO INCOMPATIBILITY]; A --> D[OTHER SYSTEM ANTIBODIES INCOMPATIBILITY]; D --> E[OTHER Rh ANTIBODIES]; D --> F[OTHER ANTIBODIES];
```

**Rh
INCOMPATIBILITY**

**ABO
INCOMPATIBILITY**

**OTHER SYSTEM
ANTIBODIES
INCOMPATIBILITY**

**OTHER Rh
ANTIBODIES**

OTHER ANTIBODIES

CAUSES OF HDN

1) Rh incompatibility

- HDN is occurred when a mother with Rh-negative blood becomes pregnant with Rh-positive baby that inherited from Rh-positive father.
- It occurs when anti-D is stimulated in mother plasma due to mother 's immune response to the antigen D on fetal's red blood cells.
- This is due to anti-D is an IgG that capable to cross placenta and hence delivered to fetal circulation.
- Rh caused HDN is less common but more severe.

2) ABO incompatibility

- HDN is arouse when a mother with blood type O becomes pregnant with a fetus with different blood types A, B or AB.
- ABO antibodies is natural occurring antibodies that clinically significant.

- 3) Others unexpected immune antibody (other than anti-D) :
- Other Rh antibodies :
 - anti-E (second most common, mild disease)
 - anti-c (third most common, mild to severe)
 - anti-C and anti-e (rare)
 - antibody combination (anti-c and anti-E occurring together, can be severe)
 - Other antibodies
 - Kell system antibodies (uncommon causes)
 - Duffy, MNSs and Kidd system antibodies (rare causes)

* Lewis and P system antibody are no occurrence in HDN.

pathogenesis



Feto maternal Hemorrhage, early maternal exposure before pX (for ABO, O-maternal blood & other for fetus)



Maternal Antibodies formed against Paternally derived antigens



In subsequent pregnancy, placental passage of maternal IgG antibodies



Maternal antibody attaches to fetal red blood cells



Fetal red blood cell hemolysis

Clinical feature of erythrobla..

- jaundice , anemia ,
- hydrops fetali,
- Hepatosplenomegaly
- splenic rupture
- pallor, Edema,
- Asphyxia
- Pulmonary hypertension
- Respiratory distress,
- Coagulopathies,
- Kernicterus,
- Hypoglycemia

Diagnosis

- Definitive diagnosis of erythroblastosis fetalis requires demonstration of;
 - blood group incompatibility and
 - corresponding antibody bound to infant's RBCs.
- ABO and Rh testing of parents,
- antibody detection (coombs test).

Treatment of Hemolytic DN/EF:

- ❖ The main goals of therapy are to
 - prevent intrauterine or extra uterine death from -severe anemia and hypoxia, and
 - avoid neurotoxicity from hyperbilirubinemia.

1. Plasmapheresis & Intrauterine blood transfusion

2. phototherapy(reduce bilirubin)

3. Exchange transfusion and

4. IV immune globulin.

➤ **Prevention:** Rh immune globulin .

Indication of Anti D:

- at 28 weeks of gestation and
 - within 72 hours of delivery
 - Bleeding (abortion, antepartum bleeding)
 - Invasive prenatal procedures
-
- Prevents the mother from producing anti-D Antibody.
 - Protect subsequent D-positive Pregnancy ,
 - but must be given each time the woman is Pregnant with D-positive fetus.

CONGÉNITAL INFECTIONS (TORCH syndrome)

- *TORCH* stands for:

T - toxoplasmosis ,

O - other

R - rubella ,

C - cytomegalovirus ,

H - herpes simplex virus

Toxoplasmosis

- caused by the protozoa *toxoplasma gondii*.
- transplacentally or during vaginal delivery.
- Overall risk of transmission to fetus is 50%.
- transmission is least in early gestation
- The most severe, if maternal infection that occurs during the first and second trimesters.

Clinical manifestations of Toxopla

- IUGR
- prematurity,
- hepatosplenomegaly,
- jaundice and
- thrombocytopenia.
- The classical **triad** consists of;
 - **hydrocephalus,**
 - **chorioretinitis and**
 - **cerebral calcification.**

Toxoplasmosis....

- ***Diagnosis :***

- Diagnosis is usually based on:
- Toxoplasma dye test
- Immunofluorescent IgG toxoplasma antibody test or IgM antibody test.

- ***Treatment :***

- A combination of pyrimethamine and sulfonamide given for one year.

Prevention :The best way to prevent congenital toxoplasmosis is by

- preventing acute maternal infection in pregnancy by;
- Avoid contact with cats ,
- Avoid eating undercooked meat in pregnancy ,
- Wash vegetables thoroughly before eating ,
- Wear gloves when handling meat and vegetables

Rubella

- 50 to 80% are infected if maternal infection occurs prior to the 8th week of gestation,
- but fetal infection is uncommon if maternal infection is in the third trimester.

Clinical manifestations of rubella

- IUGR, microcephaly, microphthalmia,
 - cataract, thrombocytopenia,
 - congenital heart disease , linear bone lesions, retinitis, convulsions and sensorineural deafness.
- The majority of manifestations however present much later in life. These include:
- Hearing loss (87%) ,
 - Congenital heart disease (46%) ,
 - Mental retardation (39%) and
 - Cataract and glaucoma (34%)

Rubella...

Diagnosis :

Viral isolation from nasopharyngeal secretions,
urine or CSF

- Detection of rubella specific IgM

Treatment :

- There is no specific treatment
- vaccination

Prevention of rubella :

- Rubella vaccine
 - Conception also should be avoided for 3 months following immunization and it is contraindicated during pregnancy.
- Avoid contact with anyone known to have acute or recent rubella infection.
- Termination of pregnancy if active maternal infection is detected

Cytomegalovirus (CMV)

- CMV is a member of the herpes group of viruses.
- Mode of transmission include:
- transplacentally ,through genital secretions , breast milk and blood transfusion
- produces intranuclear and intracytoplasmic inclusion bodies;
- enlarged cell size (cytomegalic).

CMV....

Clinical manifestations

- In about 40% asymptomatic.
- IUGR, jaundice, hepatosplenomegaly,
- petechial rash, chorioretinitis, pneumonia,
- periventricular calcifications and microcephaly.
- **the most common sensory neural hearing loss.**

Diagnosis : Isolation of virus from the urine or saliva

- CMV specific IgM antibodies

Treatment : There is no specific therapy.

- The use of ganciclovir is being studied.

Herpes simplex virus (HSV)

- commonly due to HSV type 2 (75 %)
- Modes of infection include;
 - Mostly from maternal genital tract infection
 - intrauterine infection (5%)
 - Postpartum (10%) ,
 - Transmission from a primary lesion is about 50%
 - Transmission from recurrent lesions is only 1-3%.

HSV.....

- ***Clinical manifestations***
 1. **erythematous based vesicles**
 2. Skin, eye or mouth (lesion)
 3. Encephalitic form
- Usually of later onset (3 wks)
- Has a lower incidence of viremia and vesicles

HSV....

Diagnosis :

- cytologic examination,
- direct viral culture of skin, mouth or eye lesions
- analysis of IgM specific antibody response

Treatment and prevention

- acyclovir for 14 -21 days.
- Delivery by cesarean section may reduce the risk of neonatal infection.

Congenital syphilis

- Syphilis is chronic, generalized infectious disease
- caused by spirochete *treponema pallidum*.
- Syphilis during pregnancy has a transmission rate approaching 100%.
- Fetal and perinatal deaths occur in 40%.

Clinical manifestations of C.syphilis

Early congenital syphilis

- appears during the first 2 years of life.
- two-thirds asymptomatic at birth
- hepatosplenomegaly, lymphadenopathy,
- hemolytic anemia, thrombocytopenia and
- mucocutaneous rash.
- Rhinitis (snuffles) and condylomatous lesions
- CNS abnormalities, failure to thrive, chorioretinitis,
- nephritis and nephrotic syndrome may also be seen.
- osteochondritis ,Periostitis of the long bones.

C/M of Late congenital syphilis

- frontal bossing, thickening of the sternoclavicular portion of the clavicle
- an anterior bowing of the mid portion of the tibia (Saber shins).
- synovitis involving the lower extremities
- Saddle nose is a result of rhinitis.
- **Hutchinson's triad** includes:
 - **Hutchinson's teeth, 8th nerve deafness and , interstitial keratitis .**

Diagnosis C.SYPHILIS

- VDRL (Venereal Disease Research Laboratory) ,
- RPR (Rapid Plasma Reagin)
- TPI (Treponema Pallidum Immobilization test)
- FTA-ABs (Fluorescent Treponemal Antibody Absorption test)

Treatments of C.syphilis

- treat infant whose mother had ;
 - untreated or inadequately treated syphilis or
 - whose treatment status is unknown.
- Inadequate treatment means she was;
 - treated ≤ 30 days before delivery or
 - treated with non penicillin regimen.
- Crystalline penicillin (100,000-150,000 IU/kg/d iv)
or
- Procaine penicillin (50,000 IU/kg/d Im) for 10-14 days.

Infants of diabetic mother

- Diabetic mothers have a high incidence of
 - polyhydramnios,
 - preeclampsia,
 - pyelonephritis,
 - preterm labor, and
 - chronic hypertension;
- fetal mortality rate is high especially after 32 wk
- Fetal loss throughout pregnancy is associated with poorly controlled maternal diabetes
- Macrosomia

Infants of diabetic mother...

- maternal hyperglycemia causes **fetal hyperglycemia, and fetal hyperinsulinemia** then cause;
- increased hepatic glucose uptake and glycogen synthesis, accelerated lipogenesis, and augmented protein synthesis.
- hypertrophy and hyperplasia of the pancreatic islets
increased weight of the placenta and infant organs except for the brain.
- myocardial hypertrophy, increased amount of cytoplasm in liver cells, and
- extramedullary hematopoiesis.

Infants of diabetic mother....

- Hyperinsulinism and hyperglycemia produce fetal acidosis,
 - which may increased rate of stillbirth.
- Separation of the placenta at birth leads to;
 - hyperinsulinism,
 - hypoglycemia and
 - lipolysis develop during the first hours after birth.

C/M of Infants of diabetic mother

- Hypoglycemia(about 25-50%)
- The infants tend to be jumpy, tremulous, and hyperexcitable during the 1st 3 days of life,
- although hypotonia, lethargy, and poor sucking
- Tachypnea during the 1st 2 days of life
- higher incidence of respiratory distress syndrome
- Cardiomegaly is common (30%)
- heart failure occurs in 5-10%
- Asymmetric septal hypertrophy may occur
- Birth trauma is also a common
- Neurologic development and ossification centers tend to be immature

C/M of Infants of diabetic mother....

- hyperbilirubinemia, polycythemia,
- renal vein thrombosis,
- cardiac malformations (CHD),
- lumbosacral agenesis are most common
- neural tube defects,
- hydronephrosis,
- renal agenesis and dysplasia,
- duodenal or anorectal atresia

Treatment of Infants of diabetic mother

- should be initiated before birth by frequent prenatal evaluation
- Periconception glucose control reduce complications
- Feeding
- Glucose if hypoglycemic
- Treat complication accordingly

Prognosis:

- The subsequent incidence of diabetes mellitus in infants
- May predispose to obesity that may extend into adult life.

BIRTH TRAUMA

- **Birth trauma** is as an impairment of the neonate's body function or structure due ;
- to an adverse event that occurred at birth.
- Injury may occur during labor, delivery, or after delivery,
- The majority of birth injuries are minor
- Occasionally, fatal or permanent disability.
- mostly occur because **of inappropriate or deficient medical skills or attention.**

Risk factor for birth trauma

- Cephalopelvic disproportion,
- Macrosomia – above 4 kg,
- Very low birth weight infant
- Fetus anomalies(teratoma),
- Abnormal fetal presentation (breech, face),
- Oligohydramnios
- Polyhydramnios

Risk factor for birth trauma....

- Instrumental delivery (forceps, vacuum);
- Prolonged or precipitate labour
- prematurity; postmaturity;
- contracted pelvic Small maternal stature ,
- maternal parity >6;
- maternal age younger than 16 or > 35 years;

Caput succedaneum

- scalp swelling, typically during or shortly after birth.
- It is usually caused by pressure from the
- mother's uterus or vaginal wall during delivery
- It may cross suture lines.
- disappears within the 1st few days of life.
- associated with Moulding of the head

Caput Succadaneum



Caput succedaneum

- **Treatment**
- mostly self limiting
- No specific treatment is needed but,
- If hyperbilirubinemia (PhotoTx) and
- Blood transfusion if severe bleeding

Cephalohematoma

- It is an accumulation of blood below the baby's periosteum.
- Shows up as lumps on a baby's head,
- The lesion becomes a localized firm tense mass with a palpable rim.
- They may begin to calcify by the end of the 2nd week.



Cephalohematoma....

- Most resorbed within 2 wk-3 mo,
- Infection of the hematoma is a very rare complication.

Management:

- It usually resolves spontaneously,
- phototherapy ,
- Vitamin K 1 mg IM is given.

Diagnosis and Differential Diagnosis

Cephalhaematoma

Develops hours or days after birth.

Localised haematoma to one bone limited by sutures at its edges.

Well-defined edges.

Elastic, does not pit on pressure.

Disappears within few weeks.

Caput Succedaneum

Present at birth.

Diffuse tissue oedema overlying more than one bone.

Ill-defined edges.

Soft, pits on pressure.

Disappears within 1-2 days.

Intracranial hemorrhage

Subgaleal hemorrhage

- Is a collection of blood beneath the aponeurosis that covers the scalp.
- There is often an association with vacuum-assisted delivery.
- The mechanism of injury
- linear **skull fracture**, **suture diastasis** or
- fragmentation of the superior margin of the **parietal bone**, and/or
- rupture of the emissary vein

Subgaleal hemorrhage

- .Extensive subgaleal bleeding is occasionally;
- secondary to a hemophilia.
- manifests as a firm fluctuant mass that increases in size after birth.

Subconjunctival hemorrhage

- Is bleeding that occurs when small blood vessels in the baby's eyes break.
- appears as a **bright red** band surrounding iris.
- Do not cause permanent damage to the eyes.
- The lesions resolve rapidly within the 1st 2 wk

Subconjunctival hemorrhage



Massive subdural hemorrhage

- Rare
- Usually Tears of the tentorium or falx cerebri
- rapidly deteriorate and may be fatal.
- The majority resolve without intervention.

Subarachnoid hemorrhage (SAH)

- Is rare and typically is clinically silent.
- The anastomoses between the penetrating leptomeningeal arteries or the bridging veins
- Dx - an elevated number of red blood cells in a **lumbar puncture sample.**

Managements of Intracranial hemorrhage

- Oxygen therapy
- Hemostatics: 1% vit K 1mg/kg
- Hypovolemia: 5% albumine,
- plasma 5%
- glucose – 1-10 ml/kg
- Anticonvulsants (phenobarbitol)

Soft tissue injuries

Abrasions and lacerations

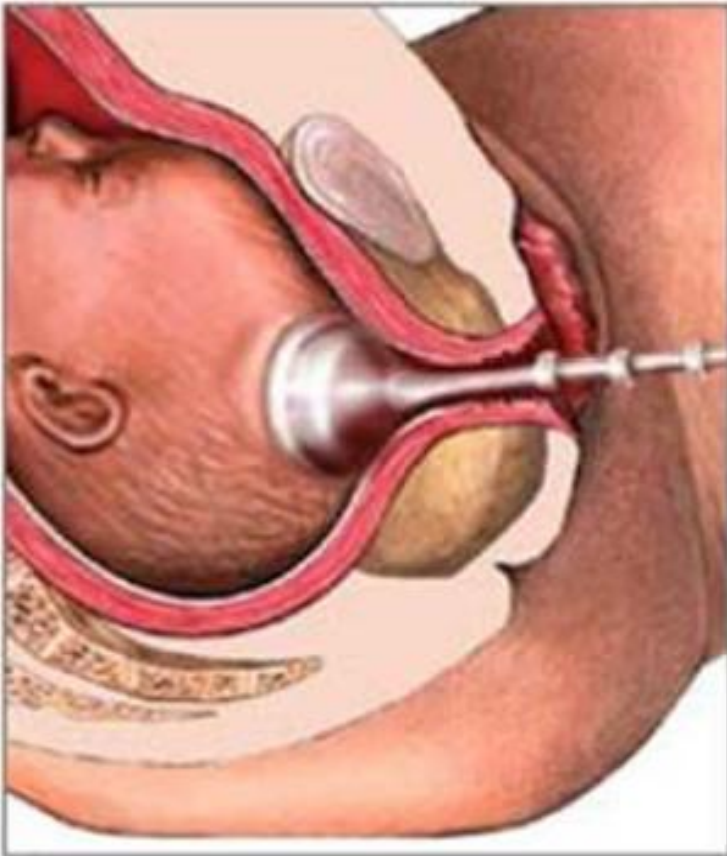
- sometimes may occur as scalpel cuts during
- cesarean delivery or during instrumental delivery (i.e, vacuum, forceps).
- Infection remains a risk.

Management consists of careful cleaning,

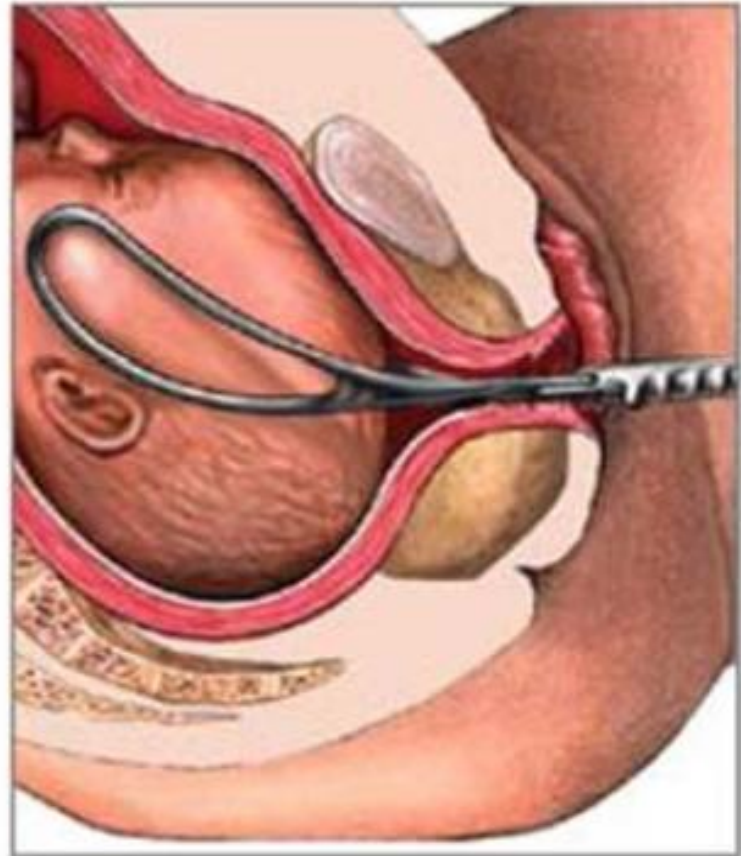
- application of antibiotic ointment, and
- observation.
- Lacerations occasionally require suturing.

Instrumental trauma

Vacuumextraction



Forceps - BREUS



Cranial nerve and spinal cord injuries

Facial paralysis

- can be caused by pressure on the facial nerves
- The affected side of the face droops and
- the infant is unable to close the eye tightly on that side.
- When crying the mouth is pulled across to the normal side.
- drooling of milk or formula from the paralyzed side of mouth.

Facial palsy



Facial palsy

- Prognosis :Generally good
- recovery usually occurs within the first month
- Surgery reserved for severe
- provide supportive treatment
- Protect cornea with moisturizing drops
- Surgical repair if not recover up to 3 months.

Spinal cord injury

- The mechanism of injury to the spinal cord
- application of strong traction when the spine is hyperextended or
- when the direction of pull is lateral.
- It may also occur with a forceful longitudinal pull when the head is firmly engaged.
- The most common part of the spinal cord involved is the **upper cervical C4** with cephalic presentation.

Spinal cord injury

- ✓ failure to establish adequate respiratory function,
- ✓ the baby usually is posing as frog,
- ✓ “oscillation” symptom is positive

DX- US & MRI

- The treatment is supportive like ;
- endotracheal intubation

3. Peripheral nerve injuries

Brachial plexus palsy:

- This result from traction to the cervical nerve roots. They may occur after breech delivery or with shoulder dystocia, with a difficult labour.

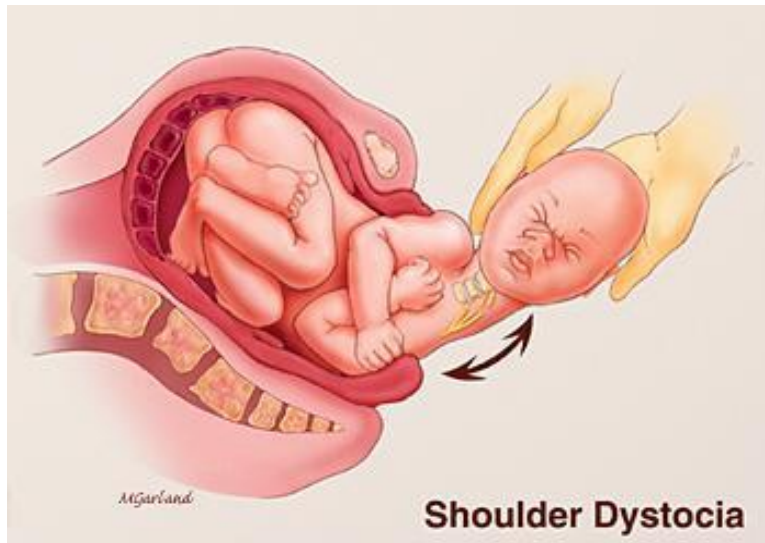


Figure 2. Classic phenotype associated with an upper brachial plexus lesion.

Spinal cord and spinal nerves

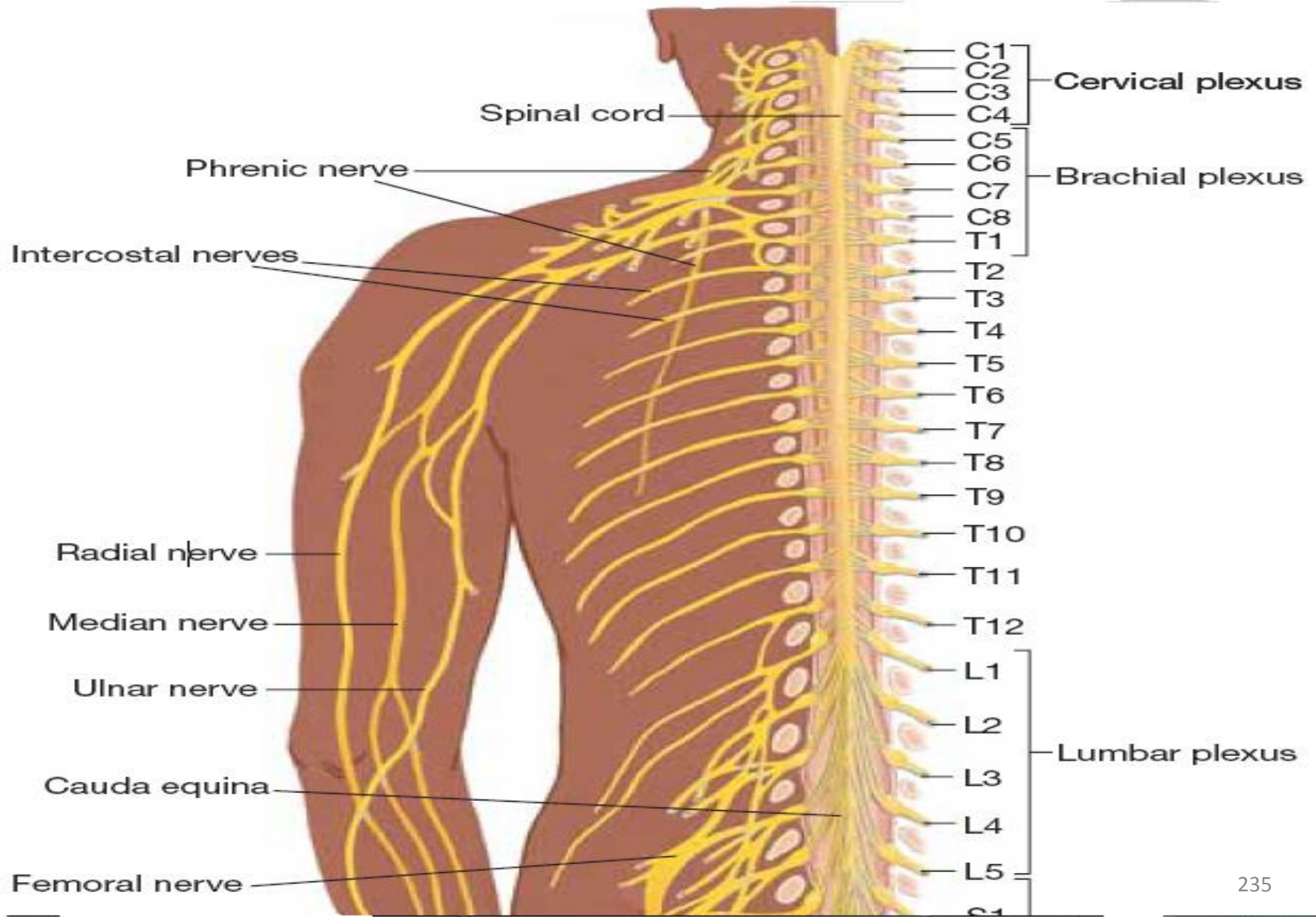


Table 8–3 **MAJOR PERIPHERAL NERVES**

Nerve	Spinal Nerves That Contribute	Distribution
Phrenic	C3–C5	• Diaphragm
Radial	C5–C8, T1	• Skin and muscles of posterior arm, forearm, and hand; thumb and first 2 fingers
Median	C5–C8, T1	• Skin and muscles of anterior arm, forearm, and hand
Ulnar	C8, T1	• Skin and muscles of medial arm, forearm, and hand; little finger and ring finger
Intercostal	T2–T12	• Intercostal muscles, abdominal muscles; skin of trunk
Femoral	L2–L4	• Skin and muscles of anterior thigh, medial leg, and foot
Sciatic	L4–S3	• Skin and muscles of posterior thigh, leg and foot

Brachial Plexus Palsy.....

- Three forms - depending on site and extent of trauma
 - **Erb-Duchenne palsy** : C 5& C6 - upper arm Palsy
 - most common form
 - **Klumpke** - C 8,T1) - lower arm palsy
 - extremely rare
 - **Kerer's paralyses** -Paralysis of entire arm
 - rare

- This baby presents with an asymmetric posture of the arms.
- The left arm is not flexed and hangs limply.
- The baby demonstrates the findings of a left-sided **ERB PARALYSIS**.



Adduction and internal rotation of the arm with pronation of the forearm



Brachial Plexus Palsy ...

Prognosis:

- If due to **edema and hemorrhage**;
 - function should *return within a few months*;
- If due to **laceration**, *permanent damage*

Brachial Plexus Palsy ;**Management**

- ✓ active and passive exercise to prevent contracture.
- ✓ Neurological/neurosurgical consultation
- ✓ If not recover within 03 months, surgical repair

Phrenic nerve paralysis

- The phrenic nerve is rarely involved;
- This nerve affects diaphragmatic function.
- Phrenic nerve injury (3rd, 4th, 5th cervical nerves) with ;
- diaphragmatic paralysis must be considered when
- cyanosis and irregular and labored respirations develop.

Phrenic nerve paralysis

- There is **no specific *treatment***:
- given oxygen if necessary.
- Spontaneous recovery is expected in 1–3 month
- After 3 months without recovery, operative intervention is indicated.

Prevention of birth injuries

A. Antenatal period

1. identification of risk factors for birth trauma &
- modification in modes of delivery.
2. CPD or malpresentation
-identified and refer the case for CS

B. Intranatal period

1. Normal delivery

- Continuous fetal monitoring with partograph to detect the early evidence of fetal distress
- Early detection of cervical arrest and prepare for C/S delivery
- Avoid frequent PV unless it is necessary .
 - Episiotomy should be carefully performed
 - The neck should not be unduly stretched

Prevention con'd

2. Special care in preterm labour

- To prevent anoxia
- To avoid strong sedation
- Liberal episiotomy and use of forceps to minimize intracranial compression

Prevention con'd

3. Forceps delivery management

- ✓ Majority of the sever injuries due to forceps
- ✓ Avoid forceps, preference for the CS
- ✓ Never apply traction unless application is a correct one

Prevention con'd

4. Vaginal breech delivery

- ✓ Care and gentleness is needed

5. To prevent intracranial injury

- ✓ Never be in hurry during delivery of the head
- ✓ Episiotomy should be done as routine to minimize head compression

Post natal

- ❖ Provide immediate new born care
- ❖ To administer Vitamin k 1 mg im
- ❖ taking history and perform neurological examination and
- ❖ new born assesment .